Machine Learning & Deep Learning

Artificial Intelligence

Jacopo Parretti

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Contents

Part I

Introduction to Machine Learning

1 What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence that enables software to use data to find solutions to specific tasks without being explicitly programmed to do so.

1.1 Machine Learning vs Statistical Modelling

In traditional statistical modelling, we follow a structured process:

1. Collect data
2. Verify and clean the data (correct or discard if not clean)
3. Use the clean data to test hypotheses, make predictions and forecasts

In contrast, Machine Learning takes a different approach:

- It is the **data** that determines which analytic techniques to use
- The computer uses data to **train** algorithms to find patterns and make predictions
- Algorithms are no longer **static** but become **dynamic**

1.2 Conventional Programming vs Machine Learning

The fundamental difference between conventional programming and machine learning can be understood through their inputs and outputs:

1.2.1 Conventional Programming

- **Input**: Program + Data
- **Output**: Result
- You write the program, give it data, and get results

1.2.2 Machine Learning

- **Input**: Data + Result
- **Output**: Program
- You give the computer data and desired results, and it learns to generate a program/algorithm

1.3 The Machine Learning Recipe

The 4 Fundamental Steps of ML

Collect data → Define models → Define objective function → Minimize error

The ML recipe consists of 4 fundamental steps:

1. Collect data

2. Define a family of possible models
3. Define an objective (error) function to quantify how well a model fits the data
4. Find the model that minimizes the error function (training/learning a model)

1.3.1 The Key Ingredients

5 Essential Ingredients

Task • Data • Model • Objective Function • Learning Algorithm

Every machine learning problem requires five essential ingredients:

- **Task**: The problem we want to solve
- **Data**: The information we use to learn
- **Model hypothesis**: The family of functions we consider
- **Objective function**: How we measure success
- **Learning algorithm**: How we find the best model

2 Machine Learning Tasks

A task represents the type of prediction being made to solve a problem on some data. We can identify a task with the set of functions that can potentially solve the problem. In general, it consists of functions assigning each **input** $x \in \mathcal{X}$ an **output** $y \in \mathcal{Y}$.

2.1 Classification Task

Definition 2.1 (Classification)**.** Find a function $f \in \mathcal{Y}^{\mathcal{X}}$ assigning each input $x \in \mathcal{X}$ a **discrete** label, $f(x) \in \mathcal{Y} = \{c_1, \ldots, c_k\}$.

In classification, the output space is a finite set of discrete categories or classes. The goal is to learn a decision boundary that separates different classes.

2.2 Regression Task

Definition 2.2 (Regression)**.** Find a function $f \in \mathcal{Y}^{\mathcal{X}}$ assigning each input $x \in \mathcal{X}$ a **continuous** label, $f(x) \in \mathcal{Y} = \mathbb{R}$.

In regression, the output is a continuous value, allowing for predictions of quantities such as prices, temperatures, or probabilities.

2.3 Density Estimation Task

Definition 2.3 (Density Estimation)**.** Find a probability distribution $f \in \Delta(\mathcal{X})$ that best fits the data $x \in \mathcal{X}$.

There is no reference to an output space (as in classification and regression tasks). Instead, we make a reasoning on the **input** x itself, trying to model the underlying probability distribution that generated the data.

2.4 Clustering Task

Definition 2.4 (Clustering)**.** Find a function $f \in \mathbb{N}^{\mathcal{X}}$ that assigns each input $x \in \mathcal{X}$ a **cluster index** $f(x) \in \mathbb{N}$.

All points mapped to the same index form a cluster. The goal is to group similar data points together without prior knowledge of the groups.

2.5 Dimensionality Reduction Task

Definition 2.5 (Dimensionality Reduction)**.** Find a function $f \in \mathcal{Y}^{\mathcal{X}}$ that **maps** each (**high-dimensional**) input $x \in \mathcal{X}$ to a **lower-dimensional** embedding $f(x) \in \mathcal{Y}$, where $\dim(\mathcal{Y}) < \dim(\mathcal{X})$.

This task aims to reduce the number of features while preserving the essential information in the data.

3 Data in Machine Learning

3.1 Data Representation

We represent inputs of our algorithm as $x \in \mathcal{X}$ and outputs as $y \in \mathcal{Y}$. The dataset is the set of both, where each input is associated with an output.

Key considerations:

- The quality of the data is crucial for the performance of the algorithm
- It is equally important to have a large amount of data to learn effective models
- Poor quality or insufficient data can lead to poor model performance

3.2 Features

Definition 3.1 (Features)**.** Features are the measurable properties or characteristics of the phenomenon being observed.

Examples:

- To identify a flower: color, petal length, petal width, sepal length, sepal width
- To identify a fruit: color, shape, size, texture, weight
- To classify images: pixel values, edges, textures, shapes

We can say that features are how our algorithm "sees" the data and are in general represented with (fixed-size) vectors.

4 Data Distributions and Learning Types

In machine learning, the information about the problem we want to solve is represented in the form of a data distribution, usually denoted as p_{data}.

4.1 Supervised Learning

4.1.1 Classification and Regression

The data distribution is over pairs of inputs and outputs:

$$p_{\text{data}} \in \Delta(\mathcal{X} \times \mathcal{Y})$$

Here:

- $\mathcal{X}$ is the input space
- $\mathcal{Y}$ is the output space (labels or targets)
- The goal is to learn a mapping from inputs to outputs using labeled data

4.2 Unsupervised Learning

4.2.1 Density Estimation, Clustering, and Dimensionality Reduction

The data distribution is only over the input space:

$$p_{\text{data}} \in \Delta(\mathcal{X})$$

Here:

- We do not have explicit labels or targets
- The goal is to discover structure, patterns, or representations within the data itself

4.3 Summary of Learning Types

Task Type	**Data Distribution**	**Learning Type**
Classification, Regression	$p_{\text{data}} \in \Delta(\mathcal{X} \times \mathcal{Y})$	Supervised Learning
Density Estimation, Clustering, Dimensionality Reduction	$p_{\text{data}} \in \Delta(\mathcal{X})$	Unsupervised Learning

Table 1: Comparison of learning types based on data distribution

Note. In **supervised learning**, each data point consists of an input and a corresponding output (label). In **unsupervised learning**, each data point consists only of the input, and the algorithm tries to find patterns or structure without explicit labels.

This distinction is fundamental in machine learning, as it determines the type of algorithms and approaches used to solve different problems.

5 Data Sampling and Dataset Splitting

5.1 The Data Distribution Problem

In both supervised and unsupervised learning, the true data distribution p_{data} is typically **unknown**. This presents a fundamental challenge:

- We do not have direct access to the underlying probability distribution that generates the data
- We only have access to a finite sample of data points drawn from this distribution

Solution: We perform **sampling** from the distribution. By collecting a representative sample of data, we can approximate the true distribution and use it to train our models.

5.2 Training, Validation, and Test Sets

To properly evaluate machine learning models, we split our data into three distinct sets. Each set serves a specific purpose in the machine learning pipeline.

Definition 5.1 (Dataset Splitting)**.** A dataset is typically divided into three subsets:

- **Training set** (D_{train}): Used to train the model
- **Validation set** (D_{val}): Used to tune hyperparameters and select the best model
- **Test set** (D_{test}): Used to evaluate the final model performance

Critical Requirement

All three sets (training, validation, test) must be sampled from the **same probability distribution**

5.3 How to Use the Three Sets

The three datasets serve different purposes in the machine learning workflow:

5.3.1 Learning/Training Phase

During the learning phase, the **training set** is used to train the machine learning algorithm, which learns patterns and relationships in the data to produce a **program** (model).

The **validation set** plays a crucial role in this phase:

- Evaluates different model configurations
- Tunes hyperparameters
- Selects the best performing model

This process involves iterative feedback between training and validation until we find the optimal configuration.

5.3.2 Testing/Model Evaluation Phase

Once the best model is selected, we use the **test set** to evaluate its performance. The trained model is applied to the test data, which it has never seen before. This gives us an unbiased estimate of how well the model will perform in real-world scenarios, telling us whether our model can truly generalize to new data.

5.4 The Importance of Distribution Similarity

5.4.1 Correct Scenario: Same Distribution

When training, validation, and test sets are sampled from the **same distribution**:

- The sets are **similar** in terms of statistical properties
- They represent the same underlying phenomenon
- However, they **do not overlap** (no data point appears in multiple sets)

This ensures that the model can generalize well from training to test data.

Example: If we're classifying fruits, all three sets might contain apples and bananas in similar proportions, but with different specific instances.

5.4.2 Incorrect Scenario: Different Distributions

When training/validation and test sets come from **different distributions**, the model learns patterns from one distribution but is tested on another. This problematic scenario typically leads to:

- Poor performance on the test set
- Unreliable predictions on new data
- Inability to generalize to real-world applications

Example: Training on apples and bananas, but testing on oranges and limes. The model cannot classify fruits it has never seen during training.

Key Principle

Training and test data must come from the **same distribution**
Violating this leads to poor generalization

6 Training and Testing: The Role of Features

6.1 Training (Learning, Induction)

During the training phase, the model learns to distinguish between different classes based on the features provided in the training data.

Example: Consider a fruit classification task with the following training data:

- *red, round, leaf, 3oz, ...* → apple
- *green, round, no leaf, 4oz, ...* → apple
- *yellow, curved, no leaf, 8oz, ...* → banana
- *green, curved, no leaf, 7oz, ...* → banana

The model learns to associate patterns in the features (color, shape, presence of leaf, weight) with the corresponding labels (apple or banana). **What distinguishes apples and bananas is based on the features!** The model identifies which feature combinations are characteristic of each class.

6.2 Testing

Once trained, the model can classify new examples based on the same features it learned during training. When presented with a new example described by the same semantic features, the model makes a prediction.

Example: Given a new fruit with features *red, round, no leaf, 4oz, ...*, the model uses the learned patterns to predict whether it is an apple or a banana.

Critical Assumption

The features of training and testing data **must be the same!**
The model can only make predictions based on features it was trained on

6.3 Learning and Generalization

Learning is fundamentally about **generalizing** from the training data. The goal is not simply to memorize the training examples, but to learn patterns that apply to new, unseen data.

However, the failure of a machine learning algorithm is often caused by a **bad selection of training samples**. The key issue is that we might introduce **unwanted correlations** from which the algorithm derives **wrong conclusions**.

Example: Consider a model trained to recognize objects in images. If all training images were captured on sunny days, the model might learn to associate sunny weather conditions with the objects. When tested on images taken on cloudy days, the model might fail to recognize the same objects because it never encountered cloudy conditions during training. The model incorrectly learned that sunny weather is a relevant feature for object recognition.

Quality and diversity of training data are crucial
Training data should be representative of all real-world conditions

7 The Complete Machine Learning Recipe

We can now revisit the complete machine learning workflow with all its essential ingredients:

7.1 The Five Ingredients of Machine Learning

Every machine learning problem requires five fundamental components that work together:

1. **Task**: The specific problem we want to solve (e.g., classify birds vs. non-birds in images)
2. **Data**: The collection of examples used to train and evaluate the model. Data quality and quantity are critical for success.
3. **Model Hypothesis**: The family of functions or models we consider as potential solutions. This defines the space of possible models the learning algorithm can explore.
4. **Objective Function**: A mathematical function that quantifies how well a model performs on the data. The learning algorithm aims to optimize this function.
5. **Learning Algorithm**: The method used to search through the model hypothesis space and find the model that optimizes the objective function.

7.2 The Interplay of Ingredients

These five ingredients are interconnected:

- The **task** determines what kind of **data** we need and what **model hypothesis** is appropriate
- The **data** influences which features are available and how we define the **objective function**
- The **model hypothesis** constrains what the **learning algorithm** can discover
- The **objective function** guides the **learning algorithm** toward better models

Understanding and carefully selecting each ingredient is essential for building effective machine learning systems.

8 Model and Hypothesis Space

8.1 What is a Model?

Definition 8.1 (Model)**.** A **model** is the implementation of a function $f \in \mathcal{F}_{\text{task}}$ that can be tractably computed.

In practice, when searching for our function f, we don't look at everything in $\mathcal{F}_{\text{task}}$. Instead, we look at a subset called the **hypothesis space**: $\mathcal{H} \subset \mathcal{F}_{\text{task}}$, which is composed of a set of models (e.g., neural networks, decision trees, linear models).

The learning algorithm seeks a **solution within the hypothesis space**. In other words, a model is a simplified representation of the world that we can actually compute and optimize.

Observation. The choice of hypothesis space is crucial:

- If $\mathcal{H}$ is too small, it may not contain a good solution
- If $\mathcal{H}$ is too large, finding the best model becomes computationally intractable
- The hypothesis space defines the types of patterns the model can learn

8.2 Types of Learning Paradigms

Three Learning Paradigms

Machine learning can be categorized into different paradigms based on the type and availability of labeled data:

Supervised Learning

In supervised learning, we have labeled data where each input is paired with its corresponding output. This includes:

- **Classification**: Predicting discrete labels
- **Regression**: Predicting continuous values

Unsupervised Learning

In unsupervised learning, we only have input data without labels. The goal is to discover structure in the data. This includes:

- **Clustering**: Grouping similar data points
- **Density Estimation**: Modeling the probability distribution of data
- **Dimensionality Reduction**: Finding lower-dimensional representations

8.2.1 Supervised Learning

In supervised learning, we have labeled data where each input is paired with its corresponding output.

8.2.2 Unsupervised Learning

In unsupervised learning, we only have input data without labels.

8.2.3 Semi-Supervised Learning

Semi-supervised learning addresses scenarios where we have a large amount of unlabeled data and only some labeled examples. This paradigm has gained significant popularity in recent years

to leverage large datasets through self-supervised learning techniques.

The key idea is to provide the model with tasks different from the one to be solved, but that allow the model to learn the data distribution from the unlabeled data. Once the model understands the general structure of the data, it can then specialize on the few labeled examples to solve the target task.

Example: Training a language model on large amounts of unlabeled text to learn general language patterns, then fine-tuning it on a small labeled dataset for sentiment analysis.

9 Model-Based vs. Instance-Based Learning

Machine learning algorithms can be further categorized based on how they make predictions:

9.1 Model-Based (Parametric) Learning

Definition 9.1 (Parametric Model)**.** A learning model that summarizes data with a **set of parameters of fixed size** (independent of the number of training examples) is called a **parametric model**.

Parametric Model Characteristics

Key characteristics:

- The model learns a fixed set of parameters from the training data
- Once trained, the original training data can be discarded
- No matter how much data you give to a parametric model, it won't change its mind about how many parameters it needs
- The model makes predictions using only the learned parameters

Examples: Naive Bayes, Linear Regression, Logistic Regression, Neural Networks

In a model-based approach, the algorithm learns a function (e.g., a decision boundary) described by parameters. For instance, a linear classifier might learn the curve $\alpha x_1 + \beta x_2 + \gamma$ that separates two classes, where α, β, γ are the parameters.

9.2 Instance-Based (Nonparametric) Learning

Definition 9.2 (Nonparametric Model)**.** These algorithms **do not learn parameters** but instead use the data itself to compare a new example through similarity metrics.

Nonparametric Model Characteristics

Key characteristics:

- The model stores (some or all of) the training data
- Predictions are made by comparing new examples to stored training examples
- The model's complexity can grow with the amount of training data
- No explicit training phase that produces a fixed set of parameters

Examples: K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Random Forest

In an instance-based approach, when a new example arrives, it is compared with nearby training examples and classified according to a similarity policy. For example, in KNN, a new point is classified based on the majority class of its k nearest neighbors in the training data.

Aspect	Model-Based	Instance-Based
Parameters	Fixed number of parameters	No fixed parameters
Training data	Can be discarded after training	Must be retained for predictions
Prediction speed	Fast (only uses parameters)	Slower (compares with stored data)
Memory	Low (stores only parameters)	High (stores training examples)
Flexibility	Fixed model complexity	Complexity grows with data

Table 2: Comparison between model-based and instance-based learning

9.3 Comparison

10 Error Function (Objective Function)

To allow an algorithm to learn, we need to provide it with a metric, a measure that helps it understand the direction of the optimal solution we are seeking. For this reason, **performance measures** are defined to enable the algorithm to learn.

10.1 What is an Error Function?

In machine learning, **training a model** means finding a **function** which maps a set of values x to a value y. We can calculate how well a predictive model is doing by comparing the **predicted values** with the **true values** for y.

Definition 10.1 (Training Error vs. Test Error)**.**

- If we apply the model to the data it was trained on, we are calculating the **training error**
- If we calculate the error on data which was **unknown** in the training phase, we are calculating the **test error**

The test error is the most important metric because it tells us how well the model generalizes to new, unseen data.

10.2 Types of Error Functions

There are different types of error functions (also called loss functions or objective functions), each suited for different tasks:

- **Mean Absolute Error (MAE)**: Measures the average absolute difference between predictions and true values
- **Mean Squared Error (MSE)**: Measures the average squared difference between predictions and true values
- **Binary Cross-Entropy**: Used for binary classification problems
- **Mean Average Precision**: Used for ranking and information retrieval tasks

The choice of error function depends on the task and the desired properties of the model.

11 Example: Polynomial Fitting

Let's consider a concrete example to understand how models, hypothesis spaces, and error functions work together.

11.1 Problem Setup

Consider a regression problem with the following setup:

- **Data**: $\mathcal{D}_n = \{(x_1, y_1), \ldots, (x_n, y_n)\}$
- Data generated from $\sin(2\pi x) + \text{noise}$
- Training set with $n = 10$ points
- **Goal**: Learn a function (shown as the purple line) that fits the data

The correct solution should capture the underlying sinusoidal pattern while being robust to the noise in the training data.

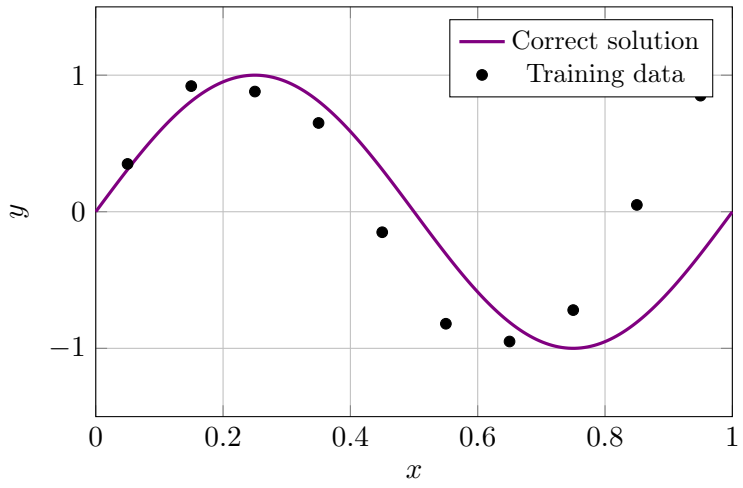


Figure 1: Polynomial Fitting Problem: Data generated from $\sin(2\pi x)$ with noise

11.2 Model: Polynomial Functions

Polynomial Model

We choose to model the data using polynomial functions:

$$f_w(x) = \sum_{j=0}^{M} w_j x^j$$

where:

- M is the **degree of the polynomial** (a hyperparameter)
- $w = \{w_0, \ldots, w_M\}$ are the **parameters** to be learned
- The **hypothesis space** for a fixed degree M is: $\mathcal{H}_M = \{f_w : w \in \mathbb{R}^{M+1}\}$

Different values of M give us different hypothesis spaces:

- $M = 0$: Constant functions (horizontal lines)
- $M = 1$: Linear functions (straight lines)
- $M = 2$: Quadratic functions (parabolas)
- $M = 3$: Cubic functions
- Higher M: More complex curves

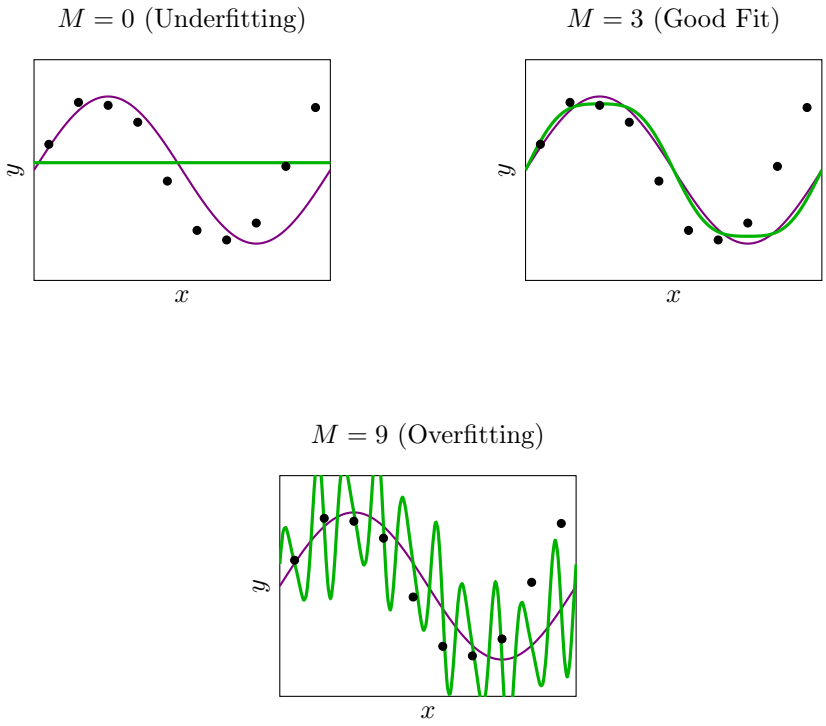


Figure 2: Polynomial fits with different degrees: $M = 0$ (underfitting), $M = 3$ (good fit), $M = 9$ (overfitting). Purple: true function, Green: fitted polynomial, Black dots: training data.

11.3 Error Function: Mean Squared Error

To measure how well our polynomial fits the data, we use the Mean Squared Error (MSE):

$$E(f; \mathcal{D}_n) = \frac{1}{n} \sum_{i=1}^{n} [f(x_i) - y_i]^2$$

This error function:

- Computes the squared difference between the prediction $f(x_i)$ and the ground-truth label y_i for each point
- Averages these squared differences over all n training points
- Is also called a **pointwise loss** because it measures the error at each individual data point

The learning algorithm's goal is to find the parameters w that minimize this error function:

$$w^* = \arg \min_{w \in \mathbb{R}^{M+1}} E(f_w; \mathcal{D}_n)$$

Choice of Model Complexity is Crucial:
- Too small M: Underfitting (cannot capture pattern)
- Appropriate M: Good fit (captures underlying pattern)
- Too large M: Overfitting (fits noise, poor generalization)

12 Underfitting and Overfitting

Two Fundamental Problems in ML

Underfitting (too simple) $\leftrightarrow$ Overfitting (too complex)

Two fundamental problems can occur when training machine learning models: underfitting and overfitting. Understanding these concepts is crucial for building models that generalize well.

12.1 Definitions

Definition 12.1 (Underfitting)**. Underfitting** is a scenario where a model is unable to capture the relationship between the input (x) and output (y) variables accurately, generating a high error rate on both the training set and unseen data (e.g., testing set).

The model is too simple and has not yet learned from the data. It performs poorly on both training and test data.

Definition 12.2 (Overfitting)**. Overfitting** occurs when the model gives accurate predictions for training data (lower training error) but not for testing data (higher testing error).

The model is too sensitive to the training data and has essentially memorized it, including its noise and peculiarities, rather than learning the underlying pattern.

12.2 The Bias-Variance Trade-off

The relationship between model complexity, training error, and test error can be visualized as follows:

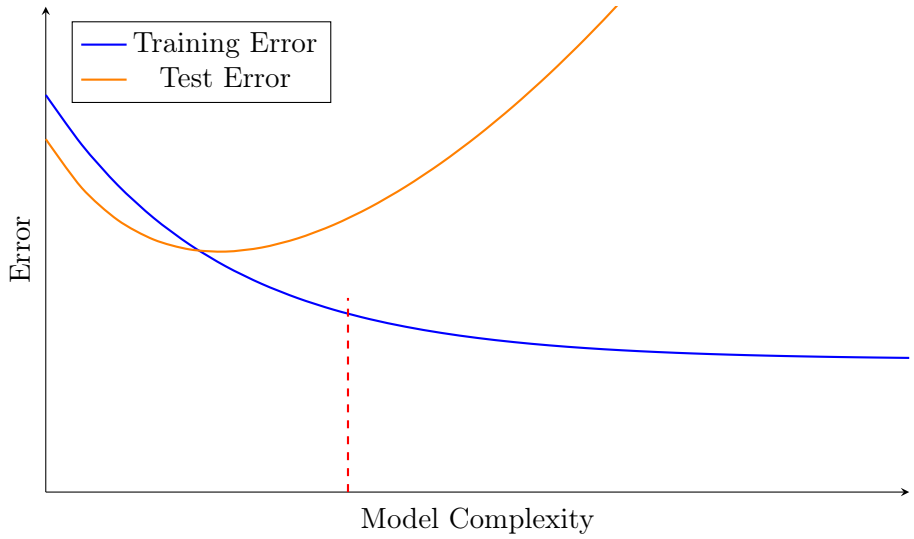


Figure 3: Bias-Variance Trade-off: Training and Test Error vs. Model Complexity

- **Left side of the graph (low complexity)**: High error rate in both training and testing $\rightarrow$ **Underfitting**
- **Middle region (optimal complexity)**: Low error rate in both training and testing $\rightarrow$ **Best Fit**
- **Right side of the graph (high complexity)**: Low error rate in training but high error rate in testing $\rightarrow$ **Overfitting**

12.3 Diagnostic Matrix

We can diagnose the state of our model by examining both training and test errors:

	Low Training Error	**High Training Error**
Low Test Error	OK (Good fit)	Underfitting
High Test Error	Overfitting	Underfitting

Table 3: Diagnostic matrix for model performance

12.4 How to Handle Overfitting

When the model is too sensitive to training data and overfits, we can apply several strategies:

1. **Try a simpler model**: Use a model with fewer parameters or lower complexity
2. **Try a less powerful model**: Choose a model architecture with reduced capacity
3. **Increase regularization impact**: Apply techniques that penalize model complexity:
   - Early stopping: Stop training before the model overfits
   - L1/L2 regularization: Add penalty terms to the loss function
4. **Use a smaller number of features**:
   - Remove some features that may be causing overfitting
   - Apply feature selection techniques to identify the most relevant features
5. **Get more data**: Increasing the training set size helps the model learn more generalizable patterns

12.5 How to Handle Underfitting

When the model has not yet learned from the data and underfits, we can try:

1. **Try more complex models**: Use models with a larger number of parameters that can capture more intricate patterns
   - Ensemble learning: Combine multiple models
2. **Less regularization**: Reduce or remove regularization constraints that may be limiting the model's capacity
3. **A larger quantity of features**: Add more features or engineer new features that better capture the underlying relationships (get more features)

12.6 Polynomial Fitting Example Revisited

Let's examine how different polynomial degrees affect the fit:

- $M = 0$ (constant function): Severe underfitting. The horizontal line cannot capture any of the sinusoidal pattern. Both training and validation errors are high.
- $M = 1$ (linear function): Still underfitting. A straight line cannot represent the curved pattern. Both errors remain high.
- $M = 3$ (cubic polynomial): Good fit. The model captures the underlying sinusoidal pattern well. Both training and validation errors are low, and the curves are similar.

- $M = 9$ (9th-degree polynomial): Overfitting. The model passes through all training points (very low training error) but oscillates wildly between them. The validation error is high because the model has learned the noise rather than the signal.

The error plot shows:

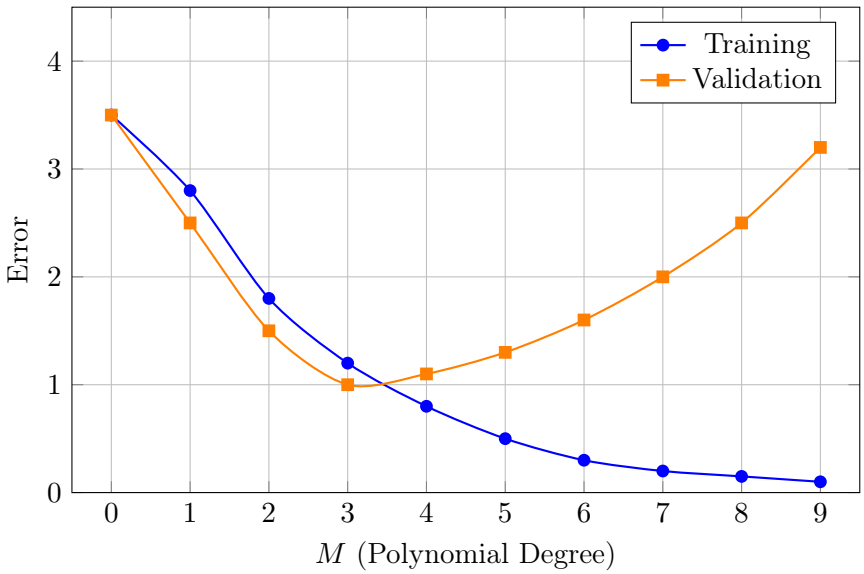


Figure 4: Training and Validation Error vs. Polynomial Degree M

- **Training error** (blue line): Decreases monotonically as M increases
- **Validation error** (orange line): Decreases initially, reaches a minimum around $M = 3$, then increases again
- The optimal model complexity is where the validation error is minimized

Key Insight

Minimizing training error alone is **not sufficient**

Must monitor validation/test error for generalization

13 Regularization

One of the Most Important Techniques in ML

Prevents overfitting and improves generalization

Regularization is one of the most important techniques in machine learning to prevent overfitting and improve model generalization.

13.1 What is Regularization?

Definition 13.1 (Regularization)**.** Regularization refers to techniques that are used to calibrate machine learning models in order to minimize the adjusted loss function and prevent overfitting or underfitting.

In practice, regularization is a modification of the training error function by appending a term $\Omega(f)$ that typically penalizes complex solutions.

13.2 The Regularized Objective Function

The regularized objective function has the following form:

$$E_{\text{reg}}(f; \mathcal{D}_n) = E(f; \mathcal{D}_n) + \lambda_n \Omega(f)$$

where:

- $E(f; \mathcal{D}_n)$ is the **training error function** (e.g., MSE)
- $\Omega(f)$ is the **regularization term** that penalizes model complexity
- λ_n is the **tradeoff hyperparameter** that controls the strength of regularization

The regularization term acts as a penalty that discourages the model from becoming too complex. By minimizing this combined objective, we find models that fit the data well while remaining relatively simple.

13.3 Effect of Regularization

The regularization term $\Omega(f)$ typically measures some notion of model complexity. When we minimize the regularized objective:

- We balance fitting the training data (low $E(f; \mathcal{D}_n)$) with keeping the model simple (low $\Omega(f)$)
- The hyperparameter λ_n controls this tradeoff
- Higher λ_n means stronger regularization (simpler models)
- Lower λ_n means weaker regularization (more complex models allowed)

13.4 Regularization in Polynomial Fitting

For the polynomial fitting example, we can regularize by penalizing polynomials with large coefficients. The regularized error function becomes:

$$E_{\text{reg}}(f_w; \mathcal{D}_n) = \frac{1}{n} \sum_{i=1}^{n} [f_w(x_i) - y_i]^2 + \frac{\lambda}{n} \|w\|^2$$

where:

- The first term is the training error (MSE)
- $\frac{\lambda}{n}\|w\|^2 = \frac{\lambda}{n} \sum_{j=0}^{M} w_j^2$ is the regularization term (L2 regularization)
- λ is the tradeoff hyperparameter

13.4.1 Effect of the Regularization Parameter λ

The choice of λ significantly affects the model:

- $\lambda \approx 10^{-18}$ **(very small)**: Almost no regularization. The model can have large coefficients and may overfit. The polynomial fits the training data very closely, including noise.
- $\lambda = 1$ **(moderate)**: Balanced regularization. The model is penalized for having large coefficients, leading to a smoother curve that generalizes better. This often provides the best tradeoff between training and validation error.

- λ **very large**: Strong regularization. The penalty for large coefficients is so high that the model becomes too simple and may underfit. The curve becomes nearly flat.

13.4.2 Regularization Path

The plot of error vs. $\ln(\lambda)$ shows:

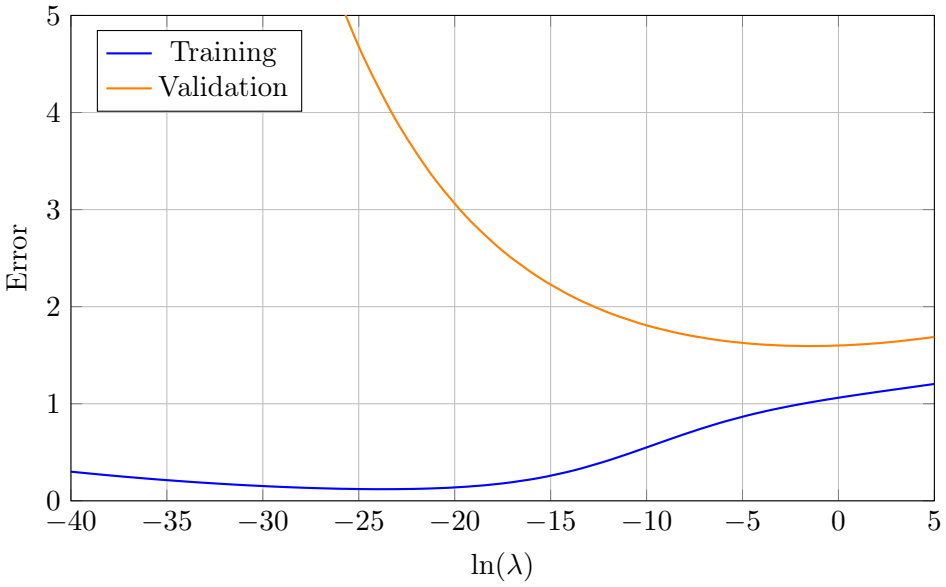


Figure 5: Training and Validation Error vs. $\ln(\lambda)$ (Regularization Strength)

- **Left side (small λ)**: Low training error, high validation error $\rightarrow$ Overfitting
- **Middle region**: Both errors are low $\rightarrow$ Good fit
- **Right side (large λ)**: Both errors increase $\rightarrow$ Underfitting

The optimal λ is found where the validation error is minimized.

13.5 Generalization and Data Size

An important theoretical result in machine learning relates training error to generalization error:

Theorem 13.1 (Generalization with Infinite Data)**.** *As the number of training samples approaches infinity, the training error approximates the generalization error:*

$$E(f; \mathcal{D}_n) \rightarrow E(f; p_{data}) \quad as \quad n \rightarrow \infty$$

This means:

- With a small dataset ($n = 15$), even a complex model ($M = 9$) may overfit because there isn't enough data to constrain the parameters
- With a large dataset ($n = 100$), the same complex model can generalize well because the abundant data prevents overfitting
- More data allows us to use more complex models without overfitting

Visual example:

- $n = 15$, $M = 9$: The 9th-degree polynomial overfits, oscillating wildly between the few training points

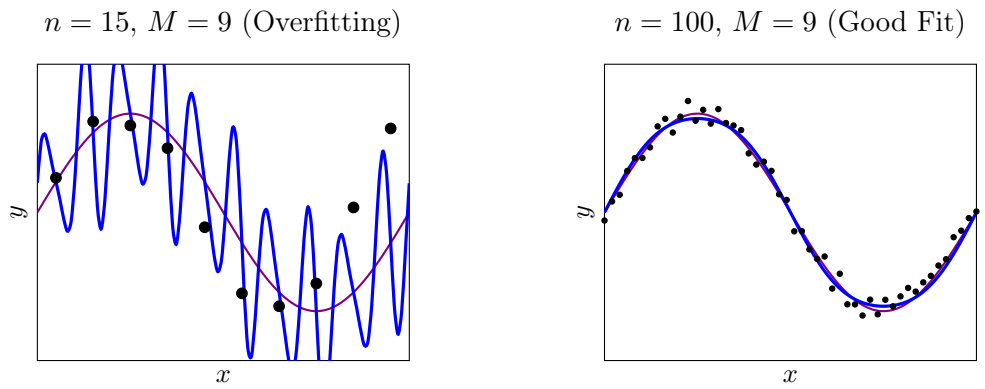


Figure 6: Effect of data size on generalization: With few data points ($n = 15$), a complex model ($M = 9$) overfits. With many data points ($n = 100$), the same model generalizes well. Purple: true function, Blue: fitted polynomial, Black dots: training data.

- $n = 100$, $M = 9$: With 100 training points, the same 9th-degree polynomial fits smoothly and generalizes well

"Get more data" is often the most effective solution to overfitting

With sufficient data, even complex models can generalize well

Part II

Model Evaluation and Regression Methods

14 Model Evaluation for Classification and Regression

14.1 Recap: Train, Validation, and Test Sets

The machine learning workflow involves three distinct datasets, each serving a specific purpose:

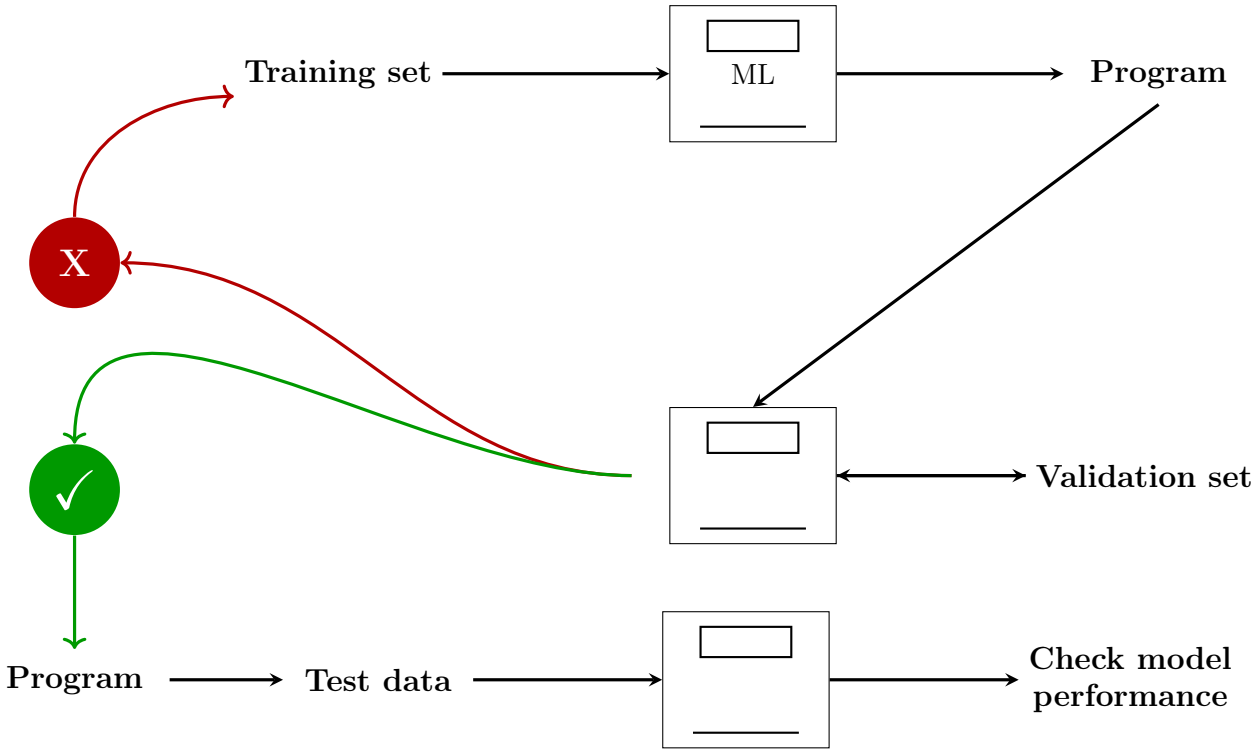


Figure 7: Machine Learning Workflow: Training set trains the model, validation set evaluates and provides feedback (thumbs down = retrain, thumbs up = proceed to testing), test set checks final performance

14.1.1 The Three Datasets

1. **Training Set**: Used to train the machine learning algorithm
   - The model learns patterns and relationships from this data
   - Parameters are optimized to minimize training error
2. **Validation Set**: Used during the learning phase
   - Also called the *mini test set*
   - Helps select the better performing algorithm among alternatives
   - Used to tune hyperparameters of the model
   - Provides feedback to adjust the model before final testing
3. **Test Set**: Used for final model evaluation

- Provides an unbiased assessment of model performance
- The model has never seen this data during training or validation
- Tells us how well the model will perform on new, real-world data

14.2 The Importance of the Validation Set

The validation set plays a crucial role in the machine learning pipeline and must be kept **separate from the training set**.

14.2.1 Why Separate the Validation Set?

The validation set is used for two main purposes:

1. **Algorithm Selection**: To pick the better performing algorithm
   - Compare different model architectures (e.g., decision trees vs. neural networks)
   - Select the model that performs best on unseen data
2. **Hyperparameter Tuning**: To decide the hyperparameters of an algorithm
   - Hyperparameters are configuration settings that are not learned from data
   - Examples: learning rate, regularization strength (λ), number of layers, tree depth
   - The validation set helps find the optimal hyperparameter values

Note. **ATTENTION**: Splitting the datasets into training and validation sets can be done randomly to avoid bias. However, there are some **specific rules** to apply when performing this split to ensure reliable model evaluation.

14.3 Specific Rules for Dataset Splitting

When splitting data into training, validation, and testing sets, you can have conflicting priorities. The key is to balance these priorities appropriately.

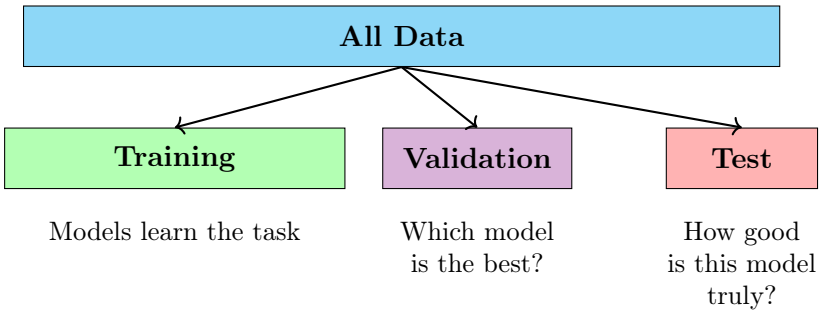


Figure 8: Dataset splitting: Training for learning, Validation for model selection, Test for final evaluation

14.3.1 Priority 1: Accurate Error Estimation

Goal: Estimate future error (i.e., validation/testing error) as accurately as possible.

How to achieve this: By making the **validation set as big as possible**.

(High confidence interval)

- A larger validation set provides more reliable error estimates

- Reduces variance in performance metrics
- Gives higher confidence that the measured performance reflects true generalization

14.3.2 Priority 2: Accurate Classifier Learning

Goal: Learn classifier as accurately as possible.

How to achieve this: By making the **training set as big as possible**.

(Better estimates, maybe better generalization)

- More training data allows the model to learn more robust patterns
- Reduces overfitting by providing diverse examples
- Generally leads to better generalization on unseen data

14.3.3 The Critical Rule

CRITICAL REQUIREMENT

Training and validation/testing instances **CANNOT OVERLAP !!!!**

- If the same data appears in both training and validation/test sets, the evaluation is invalid
- The model would be tested on data it has already seen during training
- This leads to overly optimistic performance estimates that don't reflect real-world performance
- Data leakage between sets is one of the most common mistakes in machine learning

Observation. The split between training and validation/test sets represents a fundamental trade-off:

- Larger training set → Better model learning, but less reliable error estimation
- Larger validation/test set → More reliable error estimation, but potentially worse model learning

Common splits include 70-15-15, 80-10-10, or 60-20-20 (train-validation-test), depending on the total dataset size and specific requirements.

15 Cross Validation

15.1 When Do We Need Cross Validation?

In cases where we don't have enough data to randomly split between training (e.g., 60% of the total data), validation (e.g., 20% of the total data), and testing (e.g., 20% of the total data), we need to use **cross-validation**.

Definition 15.1 (Cross Validation)**.** Cross-validation is a resampling technique that involves reusing a portion of the training set itself to validate performance by retraining not just one, but N models.

15.2 Key Characteristics of Cross Validation

- **Multiple models**: Cross-validation involves training N models instead of just one

- **Computational cost**: Performing cross-validation requires N training sessions and can be more computationally demanding
- **Better use of limited data**: When data is scarce, cross-validation allows us to use the available data more efficiently for both training and validation

15.3 Fundamental Rules of Cross Validation

Cross validation must respect certain constraints to ensure valid model evaluation:

1. **No overlap**: Training and validation cannot overlap, but $n_{\text{train}} + n_{\text{validation}} = \text{constant}$
2. **Each instance used once per fold**: At each fold (step), you use each instance only in one set, so there is no overlapping within a single fold
3. **Every sample participates**: Every sample is in both training and testing but not at the same time
    - This reduces the chances of getting a biased training set
    - Ensures all data contributes to both training and validation across different folds

15.4 K-Fold Cross Validation

K-fold cross validation is the most common form of cross validation, where the dataset is divided into k equal parts (folds).

15.4.1 How K-Fold Cross Validation Works

In the case of k-fold cross-validation:

1. We split our dataset into k groups (folds)
2. We perform training k times on $(N/k)(k-1)$ data points
3. We measure performance on the last N/k data points (the validation fold)
4. Our cross-validation performance will be the **average of the performance of the k tests**

Note. **Important constraint**: Training set and validation set cannot overlap, but the sum of training and validation data is a constant (the total dataset size).

15.4.2 Visual Representation of K-Fold Cross Validation

15.5 Example: 5-Fold Cross Validation

Let's examine a concrete example with 5 folds:

- **Step 1**: Randomly split the data into 5 folds
- **Step 2**: Test on each fold while training on 4 other folds (80% train, 20% test)
- **Step 3**: Average the results over 5 folds

15.5.1 The Process in Detail

For each of the 5 iterations:

1. **Iteration 1**: Use fold 1 as validation, folds 2-5 as training $\rightarrow$ Get Performance 1
2. **Iteration 2**: Use fold 2 as validation, folds 1,3-5 as training $\rightarrow$ Get Performance 2

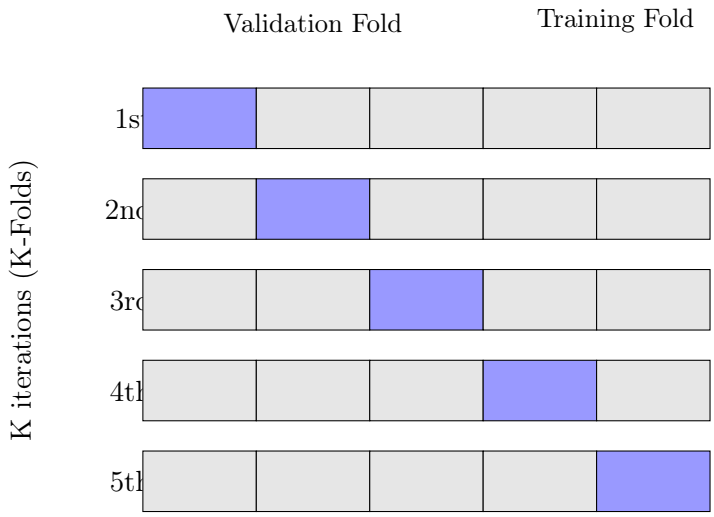


Figure 9: K-Fold Cross Validation: Each row represents one iteration where a different fold serves as validation (blue) while the rest serve as training (gray)

Figure 10: 5-Fold Cross Validation Example: Each fold serves as validation once, and the final performance is the average of all 5 iterations

3. **Iteration 3**: Use fold 3 as validation, folds 1-2,4-5 as training → Get Performance 3
4. **Iteration 4**: Use fold 4 as validation, folds 1-3,5 as training → Get Performance 4
5. **Iteration 5**: Use fold 5 as validation, folds 1-4 as training → Get Performance 5

Final Cross-Validation Performance:

$$\text{CV Performance} = \frac{1}{5}\sum_{i=1}^{5} \text{Performance}_i$$

15.6 Advantages of Cross Validation

- **Better use of limited data**: Every data point is used for both training and validation
- **More reliable performance estimates**: Averaging over multiple folds reduces variance in the performance estimate

- **Reduces bias**: Every sample participates in validation, reducing the risk of a biased validation set
- **Detects overfitting**: If performance varies significantly across folds, it may indicate overfitting or data quality issues

15.7 Disadvantages of Cross Validation

- **Computationally expensive**: Requires training k models instead of one
- **Time-consuming**: Training time is multiplied by the number of folds
- **Not suitable for very large datasets**: When data is abundant, a simple train-validation-test split is more efficient

Observation. The choice of k (number of folds) involves a tradeoff:

- **Larger** k: More accurate performance estimate, but more computationally expensive
- **Smaller** k: Faster computation, but less reliable performance estimate
- Common choices: $k = 5$ or $k = 10$
- Extreme case: $k = N$ (Leave-One-Out Cross Validation) uses each single sample as validation once

16 Leave-One-Out Cross Validation

16.1 What is Leave-One-Out Cross Validation?

Leave-One-Out Cross Validation (LOOCV) is an extreme case of k-fold cross validation that is particularly useful when we have very few data points.

Definition 16.1 (Leave-One-Out Cross Validation)**.** LOOCV is n-fold cross validation where n equals the total number of samples. In each iteration, we train on all $(n-1)$ samples and test on exactly 1 instance.

16.2 How LOOCV Works

The process involves:

1. Take the entire dataset of n samples
2. For each sample i (where $i = 1, 2, \ldots, n$):
   - Use sample i as the validation set (1 sample)
   - Use all other samples $(n-1)$ as the training set
   - Train the model and evaluate on the single validation sample
3. Repeat this process n times (once for each sample)
4. Average the performance across all n experiments

16.3 Pros and Cons of LOOCV

16.3.1 Advantages

- **Best possible classifier**: The model is learned from $n-1$ training examples, which is the maximum possible training data

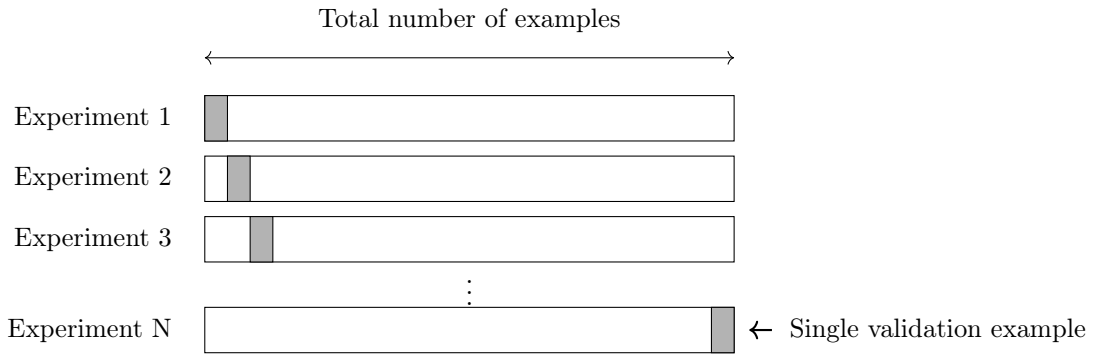


Figure 11: Leave-One-Out Cross Validation: Each experiment uses exactly one sample for validation (gray) and all others for training (white)

- **Maximizes training data**: Each model uses almost all available data for training
- **Unbiased estimate**: Provides a nearly unbiased estimate of model performance
- **Deterministic**: No randomness in the splitting process (every sample is used exactly once as validation)

16.3.2 Disadvantages

- **High computational cost**: Must re-learn everything for n times, where n is the total number of samples
- **Time-consuming**: For large datasets, LOOCV becomes impractical
- **High variance**: Performance estimates can have high variance, especially with small datasets

Note. LOOCV is most appropriate when:

- The dataset is very small (e.g., $n < 50$)
- Computational resources are available
- You need the most accurate performance estimate possible

For larger datasets, standard k-fold cross validation (with $k = 5$ or $k = 10$) is typically preferred due to its better balance between computational cost and performance estimation accuracy.

17 Stratification in Cross Validation

17.1 The Problem: Imbalanced Classes

In many real-world datasets, class labels are not evenly distributed. For example, in a medical diagnosis dataset, healthy patients might vastly outnumber sick patients. When performing cross validation on such imbalanced datasets, random splitting can lead to folds with very different class distributions.

17.2 What is Stratification?

Definition 17.1 (Stratification)**.** Stratification is a technique to keep class labels **balanced across training and validation sets** during cross validation. Instead of randomly dividing the dataset, we ensure that each fold maintains approximately the same proportion of samples for each class as the complete dataset.

17.3 How Stratified K-Fold Works

The stratification process works as follows:

1. **Divide by class**: Instead of taking the dataset and dividing it randomly into K parts, take the dataset and divide it into individual classes
2. **Split each class**: Then for each class, divide the instances into K parts
3. **Assemble folds**: Assemble the i-th part from all classes to make the i-th fold
4. **Important**: It is still random! The splitting within each class is random, but the proportions are maintained

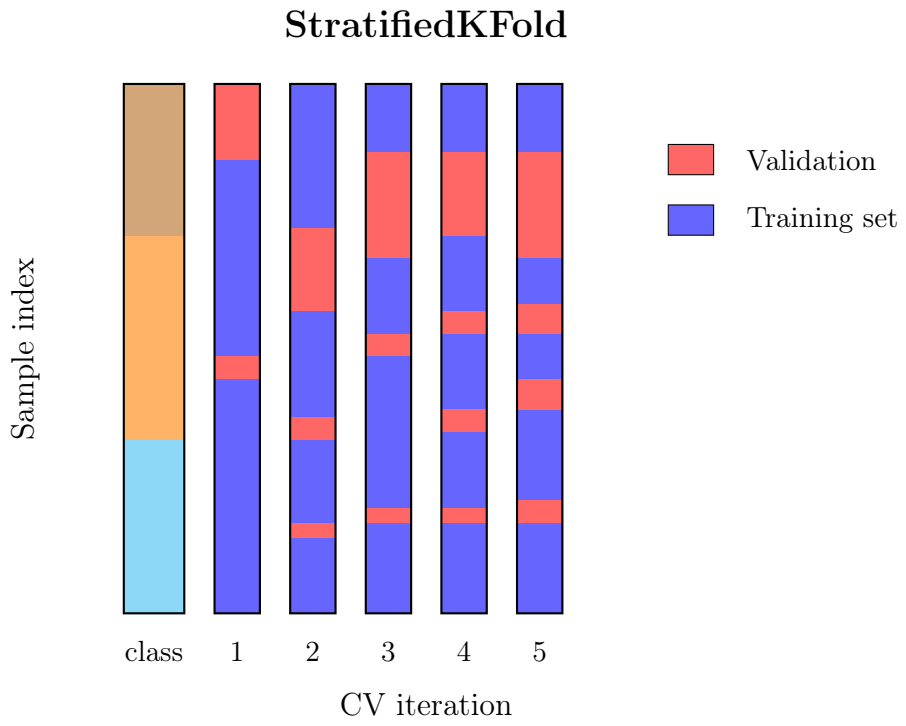


Figure 12: Stratified K-Fold: Each fold maintains the same class proportions across all folds

17.4 Example: Stratified 5-Fold Cross Validation

Let's consider a concrete scenario with an imbalanced dataset containing 300 samples, divided into three classes (A, B, and C), where Class A has more samples than the other two.

17.4.1 Imbalanced Dataset

- **Class A**: 200 samples (66.7%)
- **Class B**: 50 samples (16.7%)
- **Class C**: 50 samples (16.7%)
- **Total**: 300 samples

17.4.2 Split the Dataset into Five Folds with Stratification

Each fold will have 60 samples total, maintaining the class proportions:

- **Fold 1**: 60 samples (Class A: 40, Class B: 10, Class C: 10)

- **Fold 2**: 60 samples (Class A: 40, Class B: 10, Class C: 10)
- **Fold 3**: 60 samples (Class A: 40, Class B: 10, Class C: 10)
- **Fold 4**: 60 samples (Class A: 40, Class B: 10, Class C: 10)
- **Fold 5**: 60 samples (Class A: 40, Class B: 10, Class C: 10)

Notice that each fold maintains the same 66.7%-16.7%-16.7% class distribution as the original dataset.

17.4.3 Train and Validate

The cross-validation process proceeds as follows:

1. **Iteration 1**: Train on Folds 1, 2, 3, and 4, Validate on Fold 5
2. **Iteration 2**: Train on Folds 2, 3, 4, and 5, Validate on Fold 1
3. **Iteration 3**: Train on Folds 1, 3, 4, and 5, Validate on Fold 2
4. **Iteration 4**: Train on Folds 1, 2, 4, and 5, Validate on Fold 3
5. **Iteration 5**: Train on Folds 1, 2, 3, and 5, Validate on Fold 4

17.4.4 Average the Performance Metrics

- Calculate performance metrics for each iteration
- Average these metrics to evaluate the model's performance robustly

17.5 Why Stratification Matters

- **Prevents biased folds**: Without stratification, some folds might have very few (or no) examples of minority classes
- **Consistent evaluation**: Each fold represents the overall dataset distribution, leading to more reliable performance estimates
- **Better for imbalanced datasets**: Particularly important when dealing with class imbalance
- **Reduces variance**: Performance estimates have lower variance across folds

Note. **When to use stratification**:

- Always use stratification for classification problems with imbalanced classes
- It's generally a good practice even for balanced datasets
- Most machine learning libraries (e.g., scikit-learn) provide stratified cross-validation functions
- For regression problems, stratification is not applicable (since there are no discrete classes)

Observation. Stratification ensures that:

- Each fold is a good representative of the whole dataset
- Minority classes are adequately represented in both training and validation sets
- Performance metrics are more stable and reliable across folds
- The model sees a balanced distribution of classes during each training iteration

18 Evaluation Measures

Once we have trained our models and performed cross-validation, we need to evaluate their performance. Different tasks require different evaluation metrics.

18.1 Why Do We Need Evaluation Measures?

Evaluation measures serve two primary purposes:

1. **To decide if our model is performing well**: Assess whether the model meets the requirements for the task
2. **To decide out of many models, which one performs better than the other**: Compare different models and select the best one for deployment

18.2 Evaluation Measures by Task Type

Different machine learning tasks require different evaluation approaches:

18.2.1 Classification

Question: How often does our model classify something right/wrong?

Classification metrics measure the correctness of predictions, focusing on whether the predicted class matches the actual class.

18.2.2 Regression

Question: How close is our model to what we are trying to predict?

Regression metrics measure the distance or error between predicted continuous values and actual values.

18.2.3 Clustering

Question: How well does our model describe our data?

Clustering evaluation is generally very hard because there's often no ground truth to compare against. Metrics must assess the quality of discovered patterns.

19 Evaluation Measures for Classification

19.1 Types of Classification Problems

A classification model takes a set of features as input and produces one or more classes as output.

Definition 19.1 (Classification Types)**.** Classification can be categorized into different types based on the number of classes:

- **Binary Classification**: In the case of two classes (0/1) or one vs rest (e.g., apples vs other fruits)
- **Multi-class Classification**: In the case of having N classes to choose from (e.g., cat, dog, wolf, etc.)
- **Multi-label Classification**: A problem where an example can belong to multiple classes simultaneously

19.2 The Confusion Matrix

The confusion matrix is a fundamental tool for evaluating binary classification models. It provides a complete picture of how the model performs across different types of predictions.

19.2.1 Confusion Matrix for Binary Classification

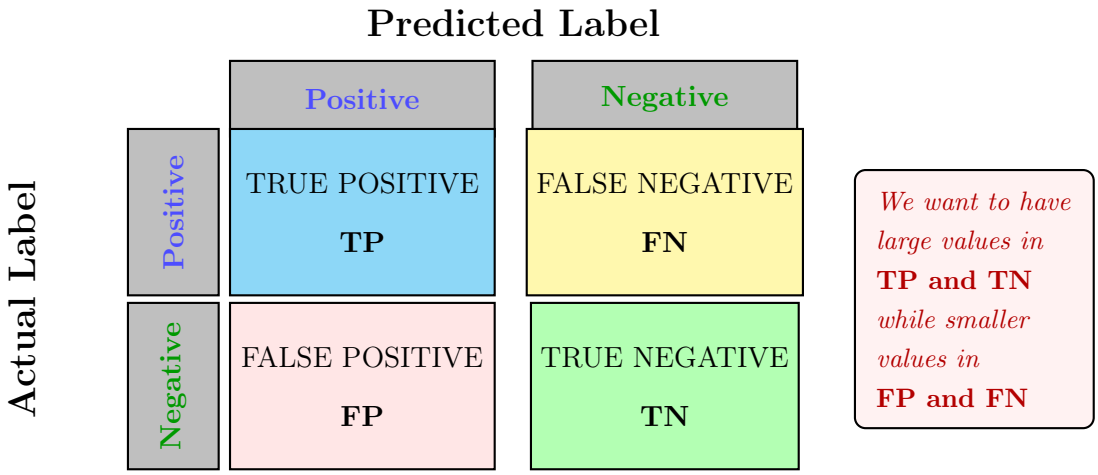


Figure 13: Confusion Matrix for Binary Classification

19.2.2 Understanding the Confusion Matrix

The confusion matrix has four components:

- **True Positive (TP)**: The model correctly predicted the positive class
  - Actual label: Positive
  - Predicted label: Positive
  - Good prediction!
- **True Negative (TN)**: The model correctly predicted the negative class
  - Actual label: Negative
  - Predicted label: Negative
  - Good prediction!
- **False Positive (FP)**: The model incorrectly predicted positive (Type I error)
  - Actual label: Negative
  - Predicted label: Positive
  - Bad prediction! (False alarm)
- **False Negative (FN)**: The model incorrectly predicted negative (Type II error)
  - Actual label: Positive
  - Predicted label: Negative
  - Bad prediction! (Missed detection)

Note. **Goal**: We want to maximize TP and TN (correct predictions) while minimizing FP and FN (incorrect predictions).

19.3 Classification Error and Accuracy

From the confusion matrix, we can derive several important metrics.

19.3.1 Classification Error

Definition 19.2 (Classification Error)**.** The classification error (also called misclassification rate) measures the proportion of incorrect predictions:

$$\text{Classification Error} = \frac{FP + FN}{TP + TN + FP + FN}$$

This represents the fraction of all predictions that were wrong.

19.3.2 Accuracy

Definition 19.3 (Accuracy)**.** Accuracy measures the proportion of correct predictions:

$$\text{Accuracy} = 1 - \text{Error} = \frac{TP + TN}{TP + TN + FP + FN}$$

This represents the fraction of all predictions that were correct.

19.4 The Problem with Accuracy: Imbalanced Classes

While accuracy is a basic measure of "goodness" of a classifier, it can be very misleading when dealing with imbalanced classes.

ATTENTION!

Accuracy is **very misleading** if we have **imbalanced classes**

19.4.1 Example: The Imbalanced Dataset Problem

Consider a dataset with severe class imbalance:

- 90 samples for class "NO"
- 10 samples for class "YES"

Naive Strategy: A trivial classifier that predicts "NO" for every sample would achieve:

$$\text{Accuracy} = \frac{90}{100} = 90\%$$

The Problem: Despite having 90% accuracy, this model is completely useless! It cannot classify any data belonging to the "YES" class. The model has learned nothing meaningful about the data.

Observation. This example illustrates why accuracy alone is insufficient for evaluating classifiers, especially with imbalanced datasets. We need additional metrics that provide a more nuanced view of model performance across all classes.

Note. When dealing with imbalanced datasets, always consider:

- Using stratified cross-validation
- Examining the confusion matrix in detail

- Using additional metrics beyond accuracy (precision, recall, F1-score)
- Considering class-specific performance
- Using techniques like oversampling, undersampling, or class weights

19.5 Misses and False Alarms

Beyond accuracy, we need more specific metrics to understand model performance. These metrics focus on different aspects of classification errors.

19.5.1 False Alarm Rate (False Positive Rate)

Definition 19.4. False Alarm Rate = False Positive Rate = $\frac{\text{FP}}{\text{FP+TN}}$

This represents the *percentage of negative samples we misclassified as positive.*

19.5.2 Miss Rate (False Negative Rate)

Definition 19.5. Miss Rate = False Negative Rate = $\frac{\text{FN}}{\text{TP+FN}}$

This represents the *percentage of positive samples we misclassified as negative.*

19.5.3 Recall (True Positive Rate)

Definition 19.6. Recall = True Positive Rate = $\frac{\text{TP}}{\text{TP+FN}}$

This represents the *percentage of positive samples we classified correctly.*

It is also known as **Sensitivity** and can be expressed as: Recall = 1 - Miss Rate

Observation. Recall answers the question: "Of all the actual positive cases, how many did we identify?"

19.5.4 Precision

Definition 19.7. Precision = $\frac{\text{TP}}{\text{TP+FP}}$

This represents the *percentage positive out of what we predicted was positive.*

Observation. Precision answers the question: "Of all the cases we predicted as positive, how many were actually positive?"

19.6 Cost of the Task

In practice, we typically do not use evaluation metrics like accuracy, recall, or precision alone. Instead, we declare a combination of them together based on the specific requirements of the task.

19.6.1 The Need for a Single Evaluation Measure

- In order to optimize a learner automatically (i.e., during training), we need a **single evaluation measure**
- How do we decide that single metric?
  - **Domain specific!** – It depends on the task

19.6.2 Weighting Errors by Cost

Depending on the **cost of the task**, we can decide whether we take care more about having:

- *Less false positives*, or
- *Less false negatives*

This is achieved through weighting them:

$$\text{Cost} = C_{\text{FP}} \cdot \text{FP} + C_{\text{FN}} \cdot \text{FN} \tag{1}$$

where C_{FP} and C_{FN} are the costs associated with false positives and false negatives, respectively.

Example 19.1. Medical Diagnosis (Cancer Detection):

- False Negative (missing a cancer diagnosis): Very high cost – patient doesn't receive treatment
- False Positive (false alarm): Lower cost – patient undergoes additional tests
- Therefore: $C_{\text{FN}} \gg C_{\text{FP}}$, we prioritize high recall

Example 19.2. Spam Email Filter:

- False Positive (marking important email as spam): High cost – user misses important messages
- False Negative (spam gets through): Lower cost – minor annoyance
- Therefore: $C_{\text{FP}} > C_{\text{FN}}$, we prioritize high precision

19.7 F-Measure (F1 Score)

The F-measure combines precision and recall into a single metric, providing a balance between the two.

Definition 19.8. The **F1 Score** is the harmonic mean of precision and recall:

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

Harmonic mean of precision and recall

One of the most frequently used metrics in machine learning

19.7.1 Properties of F1 Score

- (+) If you do some mathematics, you will see that the F1 measure is sort of an accuracy without TN
- (+) Used frequently in information retrieval systems
- (+) Provides a single number that balances precision and recall
- **Range**: $F_1 \in [0, 1]$, where 1 is perfect and 0 is the worst

Note. The harmonic mean (used in F1) is always less than or equal to the arithmetic mean. This means F1 score gives more weight to the lower value between precision and recall, ensuring both metrics need to be reasonably high for a good F1 score.

19.7.2 Generalized F-Measure

The F1 score can be generalized to F_β score, which allows adjusting the relative importance of precision and recall:

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{Precision} \times \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}} \tag{3}$$

where:

- $\beta < 1$: Emphasizes precision
- $\beta > 1$: Emphasizes recall
- $\beta = 1$: Equal weight (standard F1 score)

19.8 Multiclass Classification – Evaluation

For multiclass classification problems (more than two classes), we need to extend our evaluation metrics.

19.8.1 Multiclass Confusion Matrix

- n_{ij} is the number of examples with actual label (true label) y_i and predicted label y_j
- The **main diagonal** contains true positives for each class
- The **sum of off-diagonal elements along a column** is the number of false positives for the column label
- The **sum of off-diagonal elements along a row** is the number of false negatives for the row label

19.8.2 Multiclass Metrics

For each class i, we can compute:

$$FP_i = \sum_{j \neq i} n_{ji} \qquad FN_i = \sum_{j \neq i} n_{ij} \tag{4}$$

$$\text{Precision}_i = \frac{n_{ii}}{n_{ii} + FP_i} \qquad \text{Recall}_i = \frac{n_{ii}}{n_{ii} + FN_i} \tag{5}$$

$$\text{Multiclass Accuracy (MAcc)} = \frac{\sum_i n_{ii}}{\sum_i \sum_j n_{ij}} \tag{6}$$

19.8.3 Macro vs. Micro Averaging

When reporting overall metrics for multiclass problems, we have two main approaches:

- **Macro-averaging**: Compute metric for each class independently, then take the average

$$\text{Macro-Precision} = \frac{1}{C} \sum_{i=1}^{C} \text{Precision}_i \tag{7}$$

  This gives equal weight to each class regardless of size

- **Micro-averaging**: Aggregate the contributions of all classes to compute the average metric

$$\text{Micro-Precision} = \frac{\sum_i TP_i}{\sum_i (TP_i + FP_i)} \tag{8}$$

  This gives equal weight to each sample

19.9 ROC Curve – Receiver Operating Characteristic

The ROC curve is a powerful tool for evaluating and comparing binary classifiers, especially when the decision threshold can be varied.

19.9.1 What is the ROC Curve?

The **ROC curve** is a graphical plot that illustrates the diagnostic ability of a binary classifier as its discrimination threshold is varied.

- **X-axis**: False Positive Rate (FPR) = $\frac{FP}{FP+TN}$
- **Y-axis**: True Positive Rate (TPR) = Recall = $\frac{TP}{TP+FN}$

19.9.2 Uses of ROC Curves

The ROC curve allows us to:

- **Compare different classifiers**: Different models can be plotted on the same ROC space
- **Estimate the false positive rate / negative rate depending on the thresholds** that we set to classify
- Visualize the trade-off between sensitivity (recall) and specificity

19.9.3 Understanding the ROC Space

- **Perfect Classifier**: A point at (0, 1) – 100% TPR, 0% FPR
- **Random Classifier**: Lies on the diagonal line from (0,0) to (1,1)
- **Better Classifiers**: Curves that bow toward the upper left corner
- **Worse Classifiers**: Curves below the diagonal (worse than random guessing)

19.9.4 Area Under the ROC Curve (AUC)

Definition 19.9. The **Area Under the ROC Curve (AUC)** is a single scalar value that summarizes the performance of a classifier across all possible thresholds.

$$\text{AUC} \in [0, 1] \tag{9}$$

Goal: Maximize the Area Under the ROC Curve (AUC)

19.9.5 Interpreting AUC Values

- **AUC = 1.0**: Perfect classifier
- **AUC = 0.9 - 1.0**: Excellent
- **AUC = 0.8 - 0.9**: Good
- **AUC = 0.7 - 0.8**: Fair

- **AUC = 0.6 - 0.7**: Poor
- **AUC = 0.5**: No better than random guessing
- **AUC < 0.5**: Worse than random (predictions are inverted)

Observation. The AUC can be interpreted as the probability that the classifier will rank a randomly chosen positive instance higher than a randomly chosen negative instance.

19.9.6 Advantages of ROC and AUC

- **Threshold-independent**: Evaluates the model across all possible thresholds
- **Class imbalance robust**: Unlike accuracy, AUC is less sensitive to class imbalance
- **Single metric**: Provides a single value for model comparison
- **Visual interpretation**: The ROC curve provides intuitive visualization of classifier performance

20 Regression Evaluation Metrics

While classification tasks allow us to count correct and incorrect predictions, regression tasks require different evaluation approaches since we're predicting continuous values.

20.1 The Fundamental Difference

- **Classification**: We can count how often we are correct or wrong in our predictions
- **Regression**: We cannot do that counting!
  - Predicting y_i (not discrete, continuous value) from inputs x_i
  - The question is not "*how many times your method is wrong*" but "*how much your method is wrong with respect to the groundtruth-labels*"

Key Difference

Classification: Count errors ↔ Regression: Measure error magnitude

20.2 Mean Squared Error (MSE)

The Mean Squared Error is one of the most commonly used metrics for regression tasks.

Definition 20.1 (Mean Squared Error)**.**

$$MSE(f) = \frac{1}{N}\sum_{i=1}^{N}(f(x_i) - y_i)^2 \tag{10}$$

where:

- f is the model that takes a feature vector x as input and generates a prediction $f(x_i)$
- i, ranging from 1 to N, denotes the index of a sample in the dataset
- y_i is the ground truth (actual value)

20.2.1 Properties of MSE

- The sum of predictions minus real values over N indicates the average
- **Squaring** removes the negative sign and gives more weight to larger differences
- **The lower the MSE, the better**
- MSE is **influenced by outliers**, i.e., data values that are abnormally distant from the true regression line
- MSE is **differentiable**, which makes it suitable for gradient-based optimization algorithms

Observation. The squaring operation in MSE has two important effects:

1. It ensures all errors are positive (no cancellation between positive and negative errors)
2. It penalizes large errors more heavily than small errors (quadratic penalty)

20.3 Mean Absolute Error (MAE)

The Mean Absolute Error provides an alternative way to measure regression performance.

Definition 20.2 (Mean Absolute Error)**.**

$$MAE(f) = \frac{1}{N}\sum_{i=1}^{N}|f(x_i) - y_i| \tag{11}$$

20.3.1 Properties of MAE

- MAE is similar to MSE in that it returns absolute values of the residuals $f(x_i) - y_i$ without squaring
- It does not consider the direction of the error, which means we will not know whether negative or positive errors weigh more on the overall average
- **MAE is more robust to outliers** precisely due to the absence of squaring of distant forecast error values
- However, MAE is **not differentiable** at zero. It means that we cannot take the derivative of it, and taking derivatives is important if you want to build an algorithm that minimizes the function. Because of this, it is used a lot less than MSE.

MSE vs MAE Comparison:
- **MSE**: Differentiable, penalizes large errors more, sensitive to outliers
- **MAE**: Not differentiable at zero, treats all errors equally, robust to outliers

20.4 Choosing Between MSE and MAE

The choice between MSE and MAE depends on the specific requirements of your regression task:

20.5 Other Regression Metrics

Beyond MSE and MAE, there are other useful metrics for evaluating regression models:

Metric	Use When	Avoid When
MSE	• Large errors are particularly undesirable • You need gradient-based optimization • Outliers should be heavily penalized	• Dataset contains many outliers • All errors should be treated equally
MAE	• Outliers are present in the data • All errors should be weighted equally • You want a more interpretable metric	• You need differentiability for optimization • Large errors should be penalized more

Table 4: Guidelines for choosing between MSE and MAE

20.5.1 Root Mean Squared Error (RMSE)

$$RMSE(f) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(f(x_i) - y_i)^2} \tag{12}$$

RMSE is simply the square root of MSE. It has the advantage of being in the same units as the target variable, making it more interpretable.

20.5.2 R-Squared (R^2) Score

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - f(x_i))^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{13}$$

where $\bar{y}$ is the mean of the observed values. R^2 represents the proportion of variance in the dependent variable that is predictable from the independent variables. It ranges from 0 to 1, with 1 indicating perfect prediction.

Note. In practice, it's often beneficial to report multiple metrics to get a comprehensive view of model performance. For example, reporting both MSE (or RMSE) and MAE can provide insights into both the typical error magnitude and the presence of outliers.

Part III

Bayes Classifier and Nonparametric Classifiers

21 Bayes Classifier

21.1 Introduction

The Bayes Classifier represents a fundamental **statistical approach** for classification problems. It provides an optimal decision-making framework when all relevant probabilities are known.

Core Hypothesis

The decision problem is cast in **probabilistic terms**, where all relevant **probabilities are known**.

21.1.1 Goal of Bayes Classifier

The primary objectives are:

- **Discriminate** between different decision rules using the probabilities and their associated costs
- **Estimate** the **joint probability** $p(\mathbf{x}, y)$ from the training data set

Key Concept

The Bayes Classifier aims to estimate the joint probability distribution $p(\mathbf{x}, y)$ to make optimal classification decisions.

21.2 An Easy Example

Let's start with a simple example to understand the basic principles.

Example 21.1 (Two-Class Classification)**.** Let y be what we want to classify, and it is to be probabilistically described.

There are **two classes** y_1 and y_2 which correspond to two possible states:

a) $P(y = y_1) = 0.7$

b) $P(y = y_2) = 0.3$

These are called **a-priori** or **prior probabilities**.

Scenario: There are no measurements or observations available to help make a decision about which class y belongs to.

21.2.1 Decision Rule

Given only the prior probabilities, the optimal decision rule is straightforward:

Decision Rule:

Decide y_1 if $P(y_1) > P(y_2)$; otherwise decide y_2

Observation. Given the probabilities (70% for y_1 and 30% for y_2), the decision rule would lead to classifying y as y_1 since 0.7 is greater than 0.3.

This makes intuitive sense: without any additional information, we should choose the more probable class.

21.3 Another Example: Adding Measurements

Now let's consider a more realistic scenario where we have measurements to help us decide.

Example 21.2 (Classification with Measurements)**.** Having the previous hypothesis and additionally **a single measurement** $\mathbf{x}$, which is a random variable that depends on the class y_j, we can get:

$$p(\mathbf{x}|y_j)_{j=1,2} = \textbf{Likelihood or class-conditional probability density function}$$

i.e., *the probability of having the measurement* $\mathbf{x}$ *knowing that* y *is* y_j.

Observation. When we fix the measurement $\mathbf{x}$, the higher $p(\mathbf{x}|y_j)$ is, the more likely y_j is the correct state.

This is the key insight: measurements provide evidence that can shift our belief from the prior probabilities toward the class that better explains the observed data.

21.3.1 Visual Interpretation

The class-conditional probability density functions can be visualized as distributions over the measurement space. For two classes:

- Each class has its own probability distribution $p(\mathbf{x}|y_j)$
- These distributions may overlap in the measurement space
- Where they overlap, we need a decision rule to determine which class is more likely
- The shape and position of these distributions reflect how the measurement $\mathbf{x}$ is distributed for each class

21.4 Bayes Formula

The Bayes formula provides the mathematical framework for combining prior knowledge with observed evidence.

Bayes Formula

Assuming known $P(y_j)$ and $p(\mathbf{x}|y_j)$, the decision of ω becomes, for Bayes:

$$p(y_j, \mathbf{x}) = P(y_j|\mathbf{x})p(\mathbf{x}) = p(\mathbf{x}|y_j)P(y_j)$$

that is:

$$P(y_j|\mathbf{x}) = \frac{p(\mathbf{x}|y_j)P(y_j)}{p(\mathbf{x})} \propto p(\mathbf{x}|y_j)P(y_j)$$

where:

- $P(y_j)$ = **Prior**
- $p(\mathbf{x}|y_j)$ = **Likelihood**
- $P(y_j|\mathbf{x})$ = **Posterior**
- $p(\mathbf{x}) = \sum_{j=1}^{J} p(\mathbf{x}|y_j)P(y_j)$ = **Evidence**

21.4.1 Understanding the Components

Definition 21.1 (Bayes Formula Components)**.**

- **Prior** $P(y_j)$: Our initial belief about the probability of class y_j before seeing any data
- **Likelihood** $p(\mathbf{x}|y_j)$: The probability of observing measurement $\mathbf{x}$ given that the true class is y_j
- **Posterior** $P(y_j|\mathbf{x})$: Our updated belief about the probability of class y_j after observing measurement $\mathbf{x}$
- **Evidence** $p(\mathbf{x})$: The total probability of observing measurement $\mathbf{x}$ across all classes (acts as a normalization constant)

Observation. The evidence $p(\mathbf{x})$ is the same for all classes, which is why we can use the proportionality:

$$P(y_j|\mathbf{x}) \propto p(\mathbf{x}|y_j)P(y_j)$$

This means for classification, we only need to compare $p(\mathbf{x}|y_j)P(y_j)$ across different classes without computing the evidence explicitly.

21.4.2 Bayes Decision Rule

Optimal Bayes Decision Rule

Choose class y_j that maximizes the posterior probability:

$$\hat{y} = \arg\max_j P(y_j|\mathbf{x}) = \arg\max_j p(\mathbf{x}|y_j)P(y_j)$$

Note. The Bayes Classifier is optimal in the sense that it minimizes the probability of misclassification when all probabilities are known exactly. However, in practice, we must estimate these probabilities from data, which introduces approximation errors.

21.5 Bayes Decision Rule: Detailed Analysis

Bayes Formula Relationship

$$P(y_j|\mathbf{x}) = \frac{p(\mathbf{x}|y_j)P(y_j)}{p(\mathbf{x})} \quad \Longleftrightarrow \quad \text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}$$

21.5.1 Understanding the Posterior

- The posterior or **a-posteriori probability** is the probability that y is y_j given the observation $\mathbf{x}$
- The most important factor is the product likelihood $\times$ prior
- The evidence $p(\mathbf{x})$ is simply a scale factor, which ensures that:

$$\sum_j P(y_j|\mathbf{x}) = 1$$

Observation. Since $p(\mathbf{x})$ is the same for all classes, it acts as a normalization constant. For classification purposes, we can ignore it and simply compare the numerators $p(\mathbf{x}|y_j)P(y_j)$ across different classes.

21.5.2 The Decision Rule

From the formula of Bayes derives the **Bayes' decision rule**:

Bayes Decision Rule

Decide y_1 if $P(y_1|\mathbf{x}) > P(y_2|\mathbf{x})$, and y_2 otherwise

Equivalently, since we can ignore the evidence:

Decide y_1 if $p(\mathbf{x}|y_1)P(y_1) > p(\mathbf{x}|y_2)P(y_2)$, and y_2 otherwise

22 Naïve Bayes Classifier

The Naïve Bayes Classifier is a practical implementation of Bayes' theorem that makes a simplifying assumption about feature independence.

22.1 Core Principles

Naïve Bayes Classifier

A probabilistic classification algorithm based on **Bayes' theorem** that:

- Assumes **independence** among features
- Calculates the **posterior probability** using:

$$P(y_j|\mathbf{x}) = \frac{p(\mathbf{x}|y_j)P(y_j)}{p(\mathbf{x})}$$

22.2 The Naïve Independence Assumption

The key assumption is that features $x_1, x_2, \ldots, x_n$ are **independent** given the class label ω:

$$P(\mathbf{x}|y) = P(x_1|y) \cdot P(x_2|y) \cdots P(x_n|y) = \prod_{i=1}^{n} P(x_i|y) \tag{14}$$

Observation. This is called "naïve" because in reality, features are often correlated. However, despite this simplification, Naïve Bayes often performs surprisingly well in practice, especially for text classification tasks.

22.2.1 Training the Classifier

Key Point: Probability of each class is determined from training data.

The training process involves:

1. Estimating the prior probabilities $P(y_j)$ from the class frequencies in the training data
2. Estimating the conditional probabilities $P(x_i|y_j)$ for each feature given each class

22.3 Example: Email Spam Classification

Let's work through a complete example to understand how Naïve Bayes works in practice.

Example 22.1 (Spam Detection)**.** We want to classify an email as either:

- **Class 1**: Spam

- **Class 2**: Not spam (Ham)

We'll use a few features (words) that appear in the email.

22.3.1 Training Data

Consider the following training dataset:

Email	Contains "Free"?	Contains "Win"?	Class
E1	Yes	Yes	Spam
E2	Yes	No	Spam
E3	No	Yes	Spam
E4	No	No	Ham
E5	No	No	Ham

Table 5: Training data for spam classification

22.3.2 Step 1: Compute Class Priors

From the training data:

$$P(\text{Spam}) = \frac{3}{5} \tag{15}$$

$$P(\text{Ham}) = \frac{2}{5} \tag{16}$$

Observation. The prior probabilities reflect the class distribution in our training set: 3 out of 5 emails are spam, and 2 out of 5 are ham.

22.3.3 Step 2: Estimate Conditional Probabilities

For each feature (word) and class, we estimate how often the word appears when the email belongs to that class:

For "Free":

$$P(\text{Free} = \text{Yes}|\text{Spam}) = \frac{2}{3} \tag{17}$$

$$P(\text{Free} = \text{Yes}|\text{Ham}) = \frac{0}{2} = 0 \tag{18}$$

For "Win":

$$P(\text{Win} = \text{Yes}|\text{Spam}) = \frac{2}{3} \tag{19}$$

$$P(\text{Win} = \text{Yes}|\text{Ham}) = \frac{0}{2} = 0 \tag{20}$$

Note. Note that we also need the probabilities for the words *not* appearing:

$$P(\text{Free} = \text{No}|\text{Spam}) = 1 - \frac{2}{3} = \frac{1}{3}$$

$$P(\text{Win} = \text{No}|\text{Spam}) = 1 - \frac{2}{3} = \frac{1}{3}$$

$$P(\text{Free} = \text{No}|\text{Ham}) = 1 - 0 = 1$$

$$P(\text{Win} = \text{No}|\text{Ham}) = 1 - 0 = 1$$

22.3.4 Step 3: Classify a New Email

Validation/Testing: Consider a new email that contains "Free" but does **not** contain "Win".

We compute (ignoring the denominator (evidence) since it's the same for both classes):

$$P(\text{Spam}|\text{Free, no Win}) \propto P(\text{Spam}) \cdot P(\text{Free}|\text{Spam}) \cdot P(\text{no Win}|\text{Spam}) \tag{21}$$

$$= \frac{3}{5} \times \frac{2}{3} \times \left(1 - \frac{2}{3}\right) \tag{22}$$

$$= \frac{3}{5} \times \frac{2}{3} \times \frac{1}{3} = \frac{2}{15} \tag{23}$$

$$P(\text{Ham}|\text{Free, no Win}) \propto P(\text{Ham}) \cdot P(\text{Free}|\text{Ham}) \cdot P(\text{no Win}|\text{Ham}) \tag{24}$$

$$= \frac{2}{5} \times 0 \times 1 = 0 \tag{25}$$

Prediction: The model predicts **Spam**, because its probability is greater.

$$P(\text{Spam}|\text{Free, no Win}) = \tfrac{2}{15} > P(\text{Ham}|\text{Free, no Win}) = 0$$

22.4 The Zero Probability Problem

Observation. Notice that $P(\text{Free}|\text{Ham}) = 0$ caused the entire probability for Ham to become zero. This is a common problem in Naïve Bayes called the **zero probability problem**.

22.4.1 Solution: Laplace Smoothing

To avoid zero probabilities, we use **Laplace smoothing** (also called additive smoothing):

$$P(x_i|y_j) = \frac{\text{count}(x_i, y_j) + \alpha}{\text{count}(y_j) + \alpha \cdot n} \tag{26}$$

where:

- α is the smoothing parameter (typically $\alpha = 1$)
- n is the number of possible values for feature x_i

Note. Laplace smoothing ensures that no probability is exactly zero, which prevents the entire product from becoming zero when a feature value hasn't been observed in the training data for a particular class.

22.5 Advantages and Disadvantages

22.6 Classification Rule Summary

Naïve Bayes Classification Rule

Classify an observation $\mathbf{x}$ to class y_j if:

$$P(y_j|\mathbf{x}) = \frac{p(\mathbf{x}|y_j)P(y_j)}{p(\mathbf{x})} \text{ is maximized}$$

Advantages	Disadvantages
• Simple and fast to train and predict	• Independence assumption is often violated
• Works well with high-dimensional data	• Can be outperformed by more sophisticated models
• Requires relatively small training data	• Zero probability problem requires smoothing
• Performs well for text classification	• Cannot learn feature interactions
• Provides probabilistic predictions	• Sensitive to irrelevant features

Table 6: Advantages and disadvantages of Naïve Bayes

22.6.1 Key Points

Advantages:

- Fast to train and predict, suitable for large datasets
- Works well with high-dimensional data

Limitations:

- Independence assumption may not hold in real-world scenarios, leading to suboptimal performance
- Zero probability problem: requires techniques (like Laplace smoothing) for unseen feature-class combinations

22.7 Problems with Bayesian Classifiers

To create an optimal classifier that uses the Bayesian decision rule, you need to know:

1. **Prior probabilities** $P(y_j)$
2. **Class-conditional densities (likelihood)** $p(\mathbf{x}|y_j)$

Observation. The performance of a classifier **strongly** depends on the **goodness** of these components.

The Fundamental Problem

BUT PRACTICALLY ALL THIS INFORMATION IS NEVER AVAILABLE!!

22.7.1 What We Actually Have

More often we only have:

- A vague knowledge of the problem, from which to extract vague **a-priori probabilities**
- Some particularly representative patterns, training data, used to train the classifier (**often too few or sparse!**)

Observation.
- Estimating a-priori probabilities is usually not particularly difficult
- Estimating conditional densities (likelihood) is **more complex** → needs approximations

22.8 Challenges in Estimating Conditional Densities

Estimating conditional densities (likelihood) is more complex and needs approximations due to several challenges:

1. **High-dimensional problem**: Data can have many features → exponential combinations
   - With d features, each with k possible values, there are k^d possible feature combinations
   - This leads to the **curse of dimensionality**
2. **Zero probabilities**: Some feature combinations may never appear in training data → zero probabilities
   - This can cause the entire posterior probability to become zero
   - Requires smoothing techniques
3. **Continuous features**: Features may be continuous, requiring parametric assumptions (e.g., Gaussian)
   - We need to assume a probability distribution family
   - Then estimate the parameters of that distribution

Key Challenge:

The complexity of estimating $p(\mathbf{x}|y_j)$ grows exponentially with the number of features, making it impractical to estimate directly from data without making simplifying assumptions.

22.9 Solution: Parametric Models

One solution (out of many) is to approximate the conditional densities by **fitting a parametric model**.

Parametric Approach

We approximate the likelihood by fitting a **parametric model** (e.g., Gaussian) to the training data and estimate its parameters using, for example, **Maximum Likelihood Estimation (MLE)**.

- It is a standard way to estimate parameters associated to the likelihood from training data
- For a Gaussian model, the parameters are:

$$\boldsymbol{\theta}_j = (\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$$

  where $\boldsymbol{\mu}_j$ is the mean vector and $\boldsymbol{\Sigma}_j$ is the covariance matrix for class y_j
- The likelihood is then approximated as:

$$P(\mathbf{x}|y_j) \approx \mathcal{N}(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$$

22.9.1 Maximum Likelihood Estimation (MLE)

Definition 22.1 (Maximum Likelihood Estimation)**.** MLE chooses the parameter values that make the observed data **most probable**.

For a dataset $\mathcal{D} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_N, y_N)\}$, the likelihood function is:

$$L(\boldsymbol{\theta}_j) = \prod_{i:y_i=y_j} p(\mathbf{x}_i|y_j, \boldsymbol{\theta}_j)$$

MLE finds:

$$\hat{\boldsymbol{\theta}}_j = \arg\max_{\boldsymbol{\theta}_j} L(\boldsymbol{\theta}_j)$$

Observation. In practice, we often maximize the **log-likelihood** instead:

$$\log L(\boldsymbol{\theta}_j) = \sum_{i:y_i=y_j} \log p(\mathbf{x}_i|y_j, \boldsymbol{\theta}_j)$$

This is mathematically equivalent but numerically more stable and easier to optimize.

22.9.2 Gaussian Naïve Bayes

When features are continuous and we assume they follow a Gaussian distribution:

- For each class y_j and feature x_i, we estimate:

$$\mu_{ij} = \frac{1}{N_j} \sum_{k:y_k=y_j} x_{ki} \tag{27}$$

$$\sigma_{ij}^2 = \frac{1}{N_j} \sum_{k:y_k=y_j} (x_{ki} - \mu_{ij})^2 \tag{28}$$

  where N_j is the number of training samples in class y_j

- The likelihood for feature x_i given class y_j is:

$$p(x_i|y_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} \exp\left(-\frac{(x_i - \mu_{ij})^2}{2\sigma_{ij}^2}\right)$$

- Under the independence assumption:

$$p(\mathbf{x}|y_j) = \prod_{i=1}^{d} p(x_i|y_j)$$

Note. The Gaussian assumption is reasonable for many real-world problems, but it's important to verify whether your data actually follows this distribution. For non-Gaussian data, other parametric models (e.g., multinomial for discrete features) or non-parametric methods may be more appropriate.

22.10 Summary: From Theory to Practice

Bayesian Classification Pipeline:

1. **Theoretical Foundation**: Bayes' theorem provides the optimal decision rule
2. **Practical Challenge**: We don't have access to true probability distributions
3. **Naïve Assumption**: Assume feature independence to simplify computation
4. **Parametric Modeling**: Fit parametric models (e.g., Gaussian) to estimate likelihoods
5. **Parameter Estimation**: Use MLE to learn model parameters from training data
6. **Smoothing**: Apply techniques like Laplace smoothing to handle zero probabilities

23 MLE in Naïve Bayes: Practical Implementation

23.1 MLE for Discrete (Categorical) Features

Suppose you're training a Naïve Bayes classifier with **categorical features**. For each class y, you estimate:

- the **prior** $P(y)$ and the **likelihoods** $P(x_i|y)$

23.1.1 (a) Prior via MLE

Prior Estimation

$$P(y) = \frac{\text{number of samples with class } y}{\text{total number of samples}}$$

Observation. This is simply the relative frequency of class y in the training data. If we have 100 samples and 60 belong to class y_1, then $P(y_1) = 0.6$.

23.1.2 (b) Likelihood via MLE

For categorical features (e.g., words in text classification):

Likelihood Estimation for Categorical Features

If x_i is a categorical feature (e.g., word):

$$P(x_i = v_k|y) = \frac{\text{count of } (x_i = v_k, y)}{\text{count of } y}$$

where v_k is a specific value (category) that feature x_i can take.

Observation. This is the **maximum likelihood estimate** of the conditional probability $P(x_i = v_k|y)$. You're simply estimating probabilities by **relative frequencies**: the parameter values that **maximise** the likelihood of your training data.

Example 23.1 (Word Frequency in Spam Detection)**.** If we're classifying emails as spam or ham, and the word "free" appears in 40 out of 60 spam emails, then:

$$P(\text{free} = \text{Yes}|\text{Spam}) = \frac{40}{60} = 0.667$$

23.2 MLE for Continuous Features: Gaussian Model

If features are continuous and you assume a Gaussian likelihood:

$$P(x_i|y) = \mathcal{N}(x_i; \mu_{iy}, \sigma_{iy}^2) \quad \Rightarrow \quad P(x_i|y) = \frac{1}{\sqrt{2\pi\sigma_{iy}^2}} \exp\left(-\frac{(x_i - \mu_{iy})^2}{2\sigma_{iy}^2}\right) \tag{29}$$

Then MLE gives:

Gaussian MLE Parameters

$$\hat{\mu}_{iy} = \frac{1}{N_y} \sum_{j:y_j=y} x_{ij} \tag{30}$$

$$\hat{\sigma}^2_{iy} = \frac{1}{N_y} \sum_{j:y_j=y} (x_{ij} - \hat{\mu}_{iy})^2 \tag{31}$$

where N_y is the number of samples in class y.

Key Insight

These are just the **sample mean and variance** of that feature within each class.

23.3 Complete Example: Fruit Classification

Let's work through a complete example of Gaussian Naïve Bayes with continuous features.

Example 23.2 (Classifying Fruits by Weight)**.** We have a dataset of fruits with their weights (in grams):

Fruit	Weight (g)
Apple	150
Apple	160
Apple	155
Orange	180
Orange	190
Orange	185

Table 7: Training data for fruit classification

We want to classify a new fruit with weight $x = 158$ g.

23.3.1 Step 1: Estimate Gaussian Parameters (MLE)

For Apples:

$$\mu_{\text{Apple}} = \frac{150 + 160 + 155}{3} = \frac{465}{3} = 155 \tag{32}$$

$$\sigma^2_{\text{Apple}} = \frac{(150 - 155)^2 + (160 - 155)^2 + (155 - 155)^2}{3} \tag{33}$$

$$= \frac{25 + 25 + 0}{3} = \frac{50}{3} \approx 16.67 \tag{34}$$

For Oranges:

$$\mu_{\text{Orange}} = \frac{180 + 190 + 185}{3} = \frac{555}{3} = 185 \tag{35}$$

$$\sigma^2_{\text{Orange}} = \frac{(180 - 185)^2 + (190 - 185)^2 + (185 - 185)^2}{3} \tag{36}$$

$$= \frac{25 + 25 + 0}{3} = \frac{50}{3} \approx 16.67 \tag{37}$$

Observation. Notice that both classes have the same variance in this example. This is coincidental and won't always be the case in real data.

23.3.2 Step 2: Compute the Likelihood for a New Sample

New fruit weight: $x = 158$ g.

Using the Gaussian likelihood formula:

$$P(x|y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x-\mu_y)^2}{2\sigma_y^2}\right)$$

For Apple:

$$\begin{aligned} P(158|\text{Apple}) &= \frac{1}{\sqrt{2\pi \cdot 16.67}} \exp\left(-\frac{(158-155)^2}{2 \cdot 16.67}\right) && (38)\\ &= \frac{1}{\sqrt{104.67}} \exp\left(-\frac{9}{33.34}\right) && (39)\\ &\approx 0.097 \cdot \exp(-0.27) && (40)\\ &\approx 0.097 \cdot 0.763 && (41)\\ &\approx 0.072 && (42) \end{aligned}$$

For Orange:

$$\begin{aligned} P(158|\text{Orange}) &= \frac{1}{\sqrt{2\pi \cdot 16.67}} \exp\left(-\frac{(158-185)^2}{2 \cdot 16.67}\right) && (43)\\ &= \frac{1}{\sqrt{104.67}} \exp\left(-\frac{729}{33.34}\right) && (44)\\ &\approx 0.097 \cdot \exp(-21.87) && (45)\\ &\approx 0.097 \cdot 3.0 \times 10^{-10} && (46)\\ &\approx 1.6 \times 10^{-11} && (47) \end{aligned}$$

Observation. The likelihood for Apple is much higher than for Orange. This makes intuitive sense: 158g is much closer to the Apple mean (155g) than to the Orange mean (185g).

23.3.3 Step 3: Multiply by Class Prior

Assume equal priors (we have 3 apples and 3 oranges):

$$P(\text{Apple}) = P(\text{Orange}) = 0.5$$

Computing the posterior (proportional to likelihood $\times$ prior):

$$\begin{aligned} P(\text{Apple}|158) &\propto P(158|\text{Apple}) \cdot P(\text{Apple}) && (48)\\ &= 0.072 \cdot 0.5 && (49)\\ &\approx 0.036 && (50) \end{aligned}$$

$$P(\text{Orange}|158) \propto P(158|\text{Orange}) \cdot P(\text{Orange}) \tag{51}$$
$$= 1.6 \times 10^{-11} \cdot 0.5 \tag{52}$$
$$\approx 8 \times 10^{-12} \tag{53}$$

Prediction: Apple, because posterior probability is much higher.

$$P(\text{Apple}|158) \approx 0.036 \gg P(\text{Orange}|158) \approx 8 \times 10^{-12}$$

23.4 Key Takeaways from the Example

1. **Parameter Estimation**: We used MLE to estimate μ and σ^2 from the training data
2. **Likelihood Computation**: We computed $P(x|y)$ using the Gaussian formula with the estimated parameters
3. **Prior Incorporation**: We multiplied by the class priors (which were equal in this case)
4. **Decision**: We chose the class with the higher posterior probability
5. **Distance Matters**: The prediction makes intuitive sense—158g is much closer to the Apple distribution than the Orange distribution

Note. In practice, when computing posteriors for multiple classes, we often work with log-probabilities to avoid numerical underflow:

$$\log P(y|\mathbf{x}) = \log P(y) + \sum_{i=1}^{d} \log P(x_i|y) - \log P(\mathbf{x})$$

Since we're comparing classes, we can ignore the $\log P(\mathbf{x})$ term (it's the same for all classes).

24 Nonparametric Classifiers

24.1 Introduction to Nonparametric Methods

So far, we've discussed parametric approaches (like Gaussian Naïve Bayes) that assume a specific form for the probability distribution. Now we turn to **nonparametric methods**.

Nonparametric Methods

Given a set of examples, model the density function of the data **without making any assumptions about the form of the distribution** (e.g., Gaussian distribution).

- Estimate the probability density function **directly from the samples**
- No predetermined functional form (e.g., no assumption of Gaussian, exponential, etc.)

Observation. The term "nonparametric" is somewhat misleading—these methods often have parameters (like bin width in histograms or bandwidth in kernel density estimation), but they don't assume a specific parametric form for the underlying distribution.

Key Idea

The simplest form of nonparametric density estimation is the **histogram**.

24.2 Histogram-Based Density Estimation

The histogram is the most intuitive nonparametric density estimator.

24.2.1 How Histograms Work

Definition 24.1 (Histogram Density Estimator)**.** Dividing the sample space into a number of bins and approximate the density at the **center of each bin** by the **fraction** of points in the training data that fall into the corresponding bin:

$$p_H(x) = \frac{1}{N} \cdot \frac{\#\text{ of } x^{(k)} \text{ in same bin as } x}{\text{width of bin}}$$

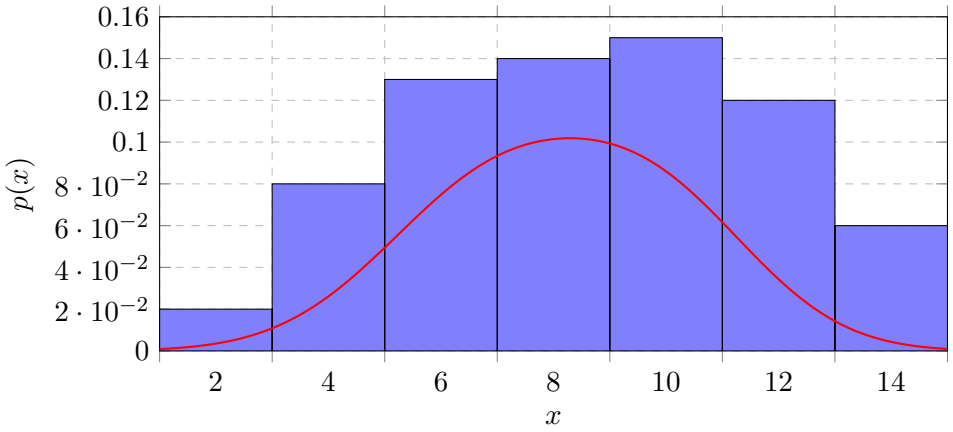
where N is the total number of training samples.

Observation. The histogram essentially counts how many data points fall into each bin and normalizes by the bin width and total number of samples to create a probability density estimate.

24.2.2 Histogram Parameters

The histogram requires two "parameters" to be defined:

1. **Bin width**: Determines the resolution of the density estimate
2. **Starting position** of the first bin: Determines where the bins are anchored



24.3 Drawbacks of Histograms

Despite their simplicity, histograms have several significant limitations:

1. **Dependence on starting position**: The density estimate depends on the starting position of the bins
   - Different starting positions can produce different density estimates
   - For multivariate data, the density estimate is also affected by the **orientation of the bins**
2. **Discontinuities**: The discontinuities of the estimate are not due to the underlying density; they are only an artifact of the chosen bin locations
   - These discontinuities make it very difficult to grasp the **structure** of the data
   - The true underlying distribution is typically smooth, but histograms create artificial jumps

3. **Curse of dimensionality**: A much more serious problem is the **curse of dimensionality**, since the number of bins grows exponentially with the number of dimensions
   - In high dimensions we would require a very large number of examples or else most of the bins would be empty
   - With d dimensions and k bins per dimension, we need k^d total bins

The Curse of Dimensionality

As the number of features or dimensions increases, the amount of data required to effectively cover the feature space increases **exponentially**.

24.4 The Curse of Dimensionality

Definition 24.2 (Curse of Dimensionality)**.** The curse of dimensionality refers to the challenges and issues that arise when working with high-dimensional data in machine learning.

As the number of features or dimensions in a dataset increases, the amount of data required to effectively cover the feature space also increases exponentially.

24.4.1 Why It Matters

- **Exponential growth**: With d dimensions, if each dimension is divided into k bins, we need k^d bins total
- **Data sparsity**: In high dimensions, data points become increasingly sparse
  - Most bins will be empty unless we have an enormous amount of data
  - The distance between points increases, making it harder to find meaningful patterns
- **Sample size requirements**: To maintain the same density of samples, we need exponentially more data as dimensions increase

Example 24.1 (Exponential Growth of Bins)**.**

- 1D: 10 bins per dimension $\rightarrow$ 10 bins total
- 2D: 10 bins per dimension $\rightarrow$ 100 bins total
- 3D: 10 bins per dimension $\rightarrow$ 1,000 bins total
- 10D: 10 bins per dimension $\rightarrow$ 10,000,000,000 bins total!

24.4.2 Mitigating the Curse of Dimensionality

To mitigate the curse of dimensionality, various techniques are employed:

- **Feature selection**: Choose only the most relevant features
- **Dimensionality reduction**: Techniques like Principal Component Analysis (PCA) reduce the number of dimensions while preserving important characteristics
- **Regularization methods**: Add constraints to prevent overfitting in high dimensions
- **Domain knowledge**: Use expert knowledge to identify which features are truly important

Practical Implication

In practice, nonparametric methods like histograms work well for 1D or 2D data, but become impractical for high-dimensional data. This motivates the development of more sophisticated techniques.

24.5 Probability Density: Mathematical Foundation

Before we proceed with more advanced nonparametric techniques, let's return to the basic definition of probability to get a solid idea of what we are trying to accomplish.

Probability in a Region

The probability that a vector $\mathbf{x}$, drawn from a distribution $P(\mathbf{x})$, will fall in a region $\mathcal{R}$ of the sample space is:

$$P = \int_{\mathbf{x}' \in \mathcal{R}} P(\mathbf{x}')\, d\mathbf{x}'$$

where:

- $\mathbf{x}$ is a particular vector we are asking about ("drawn from the distribution")
- $\mathbf{x}'$ is a dummy variable used to sum/integrate over all vectors in $\mathcal{R}$
- $\mathcal{R}$ is a region in the feature space

Observation. This integral formulation is the foundation for all density estimation methods. The key insight is that if we can estimate the probability in small regions around each point, we can reconstruct the entire probability density function.

24.5.1 Key Concepts

- **Probability density function** $P(\mathbf{x})$: Describes how probability is distributed over the feature space
- **Integration over a region**: The probability of falling in region $\mathcal{R}$ is the integral of the density over that region
- **Dummy variable** $\mathbf{x}'$: Used for integration; represents all possible points in the region
- **Normalization**: The integral over the entire space must equal 1:

$$\int_{\text{all space}} P(\mathbf{x}')\, d\mathbf{x}' = 1$$

Note. For discrete distributions, the integral becomes a sum:

$$P = \sum_{\mathbf{x}' \in \mathcal{R}} P(\mathbf{x}')$$

This is the discrete analog of the continuous case and is what we use when working with categorical data.

24.6 Binomial Distribution Approach

Now let's develop a more rigorous approach to nonparametric density estimation based on probability theory.

24.6.1 Setup

Suppose now that N vectors $\{\mathbf{x}(1), \mathbf{x}(2), \ldots, \mathbf{x}(N)\}$ are drawn **independently and identically distributed (i.i.d.)** from the distribution. The probability that k of these N vectors fall in $\mathcal{R}$ is given by the binomial distribution:

Binomial Distribution

$$P(k) = \binom{N}{k} P^k (1-P)^{N-k}$$

where:

$$\binom{N}{k} = \frac{N!}{k!(N-k)!}$$

and the expected value for k is:

$$\mathbb{E}[k] = NP$$

Observation. **Recall**: The expectation is the average value we would get if we repeated drawing samples many times.

The binomial distribution tells us how likely it is to observe exactly k points in region $\mathcal{R}$ when we draw N samples, given that each point has probability P of falling in $\mathcal{R}$.

24.6.2 Mean and Variance of the Ratio

The mean and variance of the ratio k/N are:

$$\mathbb{E}\left[\frac{k}{N}\right] = P \tag{54}$$

$$\text{Var}\left[\frac{k}{N}\right] = \mathbb{E}\left[\left(\frac{k}{N} - P\right)^2\right] = \frac{P(1-P)}{N} \tag{55}$$

Observation. Therefore, as $N \to \infty$, the distribution becomes **sharper**, the variance gets **smaller**, thus we can obtain a **better estimate** of the probability P from the mean fraction of the points that fall within $\mathcal{R}$:

$$P \cong \frac{k}{N}$$

This is the law of large numbers in action: with more samples, our estimate becomes more accurate.

24.7 Density Estimation Formula

24.7.1 Small Region Assumption

On the other hand, if we assume that $\mathcal{R}$ is **so small** that $P(\mathbf{x})$ **does not vary** appreciably within it, then:

$$\int_{\mathcal{R}} P(\mathbf{x}')\, d\mathbf{x}' \cong P(\mathbf{x})V$$

where V is the **volume enclosed** by region $\mathcal{R}$.

Observation. This approximation assumes that the probability density is approximately constant within the small region $\mathcal{R}$, so we can pull $P(\mathbf{x})$ out of the integral, leaving just the volume V.

24.7.2 Combining the Results

Merging with the previous result, we obtain:

$$P = \int_{\mathcal{R}} p(\mathbf{x}')\, d\mathbf{x}' \cong p(\mathbf{x})V \tag{56}$$

$$P \cong \frac{k}{N} \tag{57}$$

Therefore:

$$p(\mathbf{x}) \cong \frac{k}{NV}$$

Key Result

This estimate becomes more accurate as we **increase the number of sample points** N and **shrink the volume** V.

24.8 The Trade-off Problem

In practice, the value of N (the total number of examples) is fixed.

24.8.1 The Dilemma

- In order to improve the accuracy of the estimate $P(\mathbf{x})$ we could let V to approach zero, but then the region $\mathcal{R}$ would become so small that it would enclose no examples
- This means that in practice, we will have to find a **compromise value of the volume** V:
  - **Large enough** to include enough examples within $\mathcal{R}$
  - **Small enough** to support the assumption that $P(\mathbf{x})$ is constant within $\mathcal{R}$

The Fundamental Trade-off

- Too large $V \rightarrow$ Poor approximation (density varies within region)
- Too small $V \rightarrow$ Too few samples (high variance in estimate)

24.9 General Nonparametric Density Estimation Formula

General Expression for Nonparametric Density Estimation

$$P(\mathbf{x}) \cong \frac{k}{NV}$$

where:

- V is the volume surrounding $\mathbf{x}$
- N is the total number of examples
- k is the number of examples inside V

24.10 Two Approaches to Nonparametric Density Estimation

When applying this result to practical density estimation problems, two basic approaches can be adopted:

1. **Fix V and determine k from the data**
   - This leads to **kernel density estimation (KDE)**, e.g., Parzen window
   - We define a fixed-size region around each point and count how many samples fall within it
2. **Fix k and determine V from the data**
   - This gives rise to the **k-nearest-neighbor (k-NN) approach**
   - We find the k nearest neighbors and measure the volume needed to enclose them

Convergence Property

Both k-NN and KDE converge to the true probability density as $N \to \infty$, provided that:
- V shrinks with N
- k grows with N appropriately

24.10.1 Comparison of Approaches

Kernel Density Estimation (KDE)	**k-Nearest Neighbor (k-NN)**
Fix volume V	Fix number of neighbors k
Count samples k in fixed region	Adjust volume V to include k samples
Smooth density estimate	Adaptive to local density
Can be zero in sparse regions	Always uses k neighbors
Examples: Parzen window, Gaussian kernel	Examples: k-NN classifier, k-NN regressor

Table 8: Comparison of KDE and k-NN approaches

Observation. **KDE** produces smoother estimates but may miss structure in sparse regions, while **k-NN** adapts to local density but can produce less smooth estimates. The choice depends on the specific application and data characteristics.

24.10.2 Choosing Parameters

- **For KDE**: Choose bandwidth (related to V)
  - Too small: noisy, overfitting
  - Too large: oversmoothing, missing details
- **For k-NN**: Choose k
  - Too small: sensitive to noise, high variance
  - Too large: oversmoothing, high bias

Note. Both methods require careful tuning of their parameters (bandwidth for KDE, k for k-NN). Cross-validation is commonly used to select these parameters in practice.

25 k-Nearest Neighbors (k-NN)

25.1 Visual Comparison: k-NN vs. Parzen Windows

Key Difference:
- **Parzen Window**: Fixed volume V (radius R), variable k
- **k-NN**: Fixed k (number of neighbors), variable volume V (radius R)

Observation. In the visual comparison:

- **Parzen Window**: Uses a fixed-size circle (radius R) around the query point (yellow cross). The number of points inside varies depending on local density.
- **k-NN (e.g., 3-NN)**: The circle grows or shrinks to always include exactly $k = 3$ nearest neighbors. In dense regions, the circle is small; in sparse regions, it's large.

25.2 Parzen Window: Practical Considerations

When to Use Parzen Windows

Parzen window is great for **low-dimensional data** (2–3 features, maybe up to 5).

- In higher dimensions, the method becomes **computationally expensive**, and the density estimates are **unreliable** because of the curse of dimensionality
- For this reason, in A.A. 2025–2026, we focus on other methods better suited for high-dimensional data

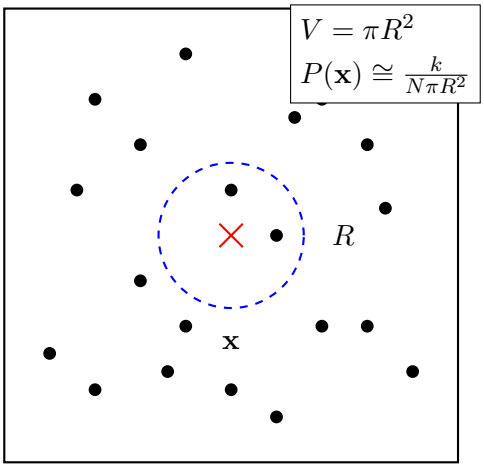
Note. While Parzen windows provide smooth density estimates, they suffer from the curse of dimensionality. As dimensions increase, the fixed window size becomes either too large (over-smoothing) or too small (too few samples), making the method impractical for modern high-dimensional datasets.

25.3 k-Nearest Neighbors Algorithm

The k-NN algorithm is one of the simplest and most intuitive machine learning algorithms.

25.3.1 Basic Principle

The volume is grown surrounding the estimation point $\mathbf{x}$, until it encloses a total of k data points.



k-NN Density Estimation

$$P(\mathbf{x}) \cong \frac{k}{NV}$$

where:
- V is the volume surrounding $\mathbf{x}$
- N is the total number of examples
- k is the number of examples inside V (fixed)

25.4 k-NN for Classification

25.4.1 1-Nearest Neighbor (1-NN)

Given an unknown point $\mathbf{x}^*$, we consider $s_i \in \{s_1, \ldots, s_N\} = S$ the nearest point (NN) of the point $\mathbf{x}^*$ if:

$$d(\mathbf{x}^*, s_i) = \min_l d(\mathbf{x}^*, s_l), \quad l = 1 \ldots N$$

where $d(\cdot)$ is a **distance measure** defined on the feature space.

Observation. If you interpret each point of set S as a prototype of a class, you can then see the equation above as a rule of classification **1-NN** associating the sample $\mathbf{x}$ to the class j to which the point s_i belongs.

Example 25.1 (Common Distance Measures).
- **Euclidean distance**: $d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^{d}(x_i - y_i)^2}$
- **Manhattan distance**: $d(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{d} |x_i - y_i|$
- **Minkowski distance**: $d(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{d} |x_i - y_i|^p\right)^{1/p}$

25.4.2 Generalizing to k-NN

Rule 1-NN can be generalised to define a k**-NN rule**, which consists of determining the k points belonging to the set S closest to the observation $\mathbf{x}$.

k-NN Classification Rule

The classification rule k-NN associates observation $\mathbf{x}$ with class i having the **largest number of elements** among the nearest k.
We call $U(\mathbf{x})$ the set of k points closer to $\mathbf{x}$.

25.4.3 Majority Voting

For example, with odd k and two classes, y_1 and y_2, the decision rule can choose the class with the most samples in $U(\mathbf{x})$. This is called **majority voting**.

Majority Voting

Choose the class that appears most frequently among the k nearest neighbors.

Example 25.2 (3-NN Classification). Suppose we have $k = 3$ and two classes (red and blue):
- Query point $\mathbf{x}^*$ has 3 nearest neighbors: 2 red, 1 blue
- Majority voting: $\mathbf{x}^*$ is classified as **red**

25.5 Choosing k

The choice of k is critical for k-NN performance:

Value of k	Advantages	Disadvantages
Small k (e.g., 1-3) - More flexible decision boundaries - High variance - Overfitting	- Captures local structure - Sensitive to noise	
Large k - Smoother decision boundaries - Lower variance - High bias - Underfitting	- More robust to noise - May miss local patterns	
Odd k	- Avoids ties in binary classification	-

Table 9: Trade-offs in choosing k

Rule of Thumb

- Start with $k = \sqrt{N}$ where N is the number of training samples
- Use cross-validation to find the optimal k
- For binary classification, use odd k to avoid ties

25.6 Advantages and Disadvantages of k-NN

Advantages	Disadvantages
• Simple and intuitive	• Computationally expensive at prediction time
• No training phase (lazy learning)	• Requires storing all training data
• Naturally handles multi-class problems	• Sensitive to irrelevant features
• Non-parametric (no assumptions about data distribution)	• Sensitive to the scale of features
• Can capture complex decision boundaries	• Curse of dimensionality
• Easy to update with new data	• Choosing optimal k can be difficult

Table 10: Advantages and disadvantages of k-NN

25.7 Practical Considerations

25.7.1 Feature Scaling

Observation. k-NN is **highly sensitive to feature scaling** because it relies on distance calculations. Features with larger ranges will dominate the distance metric.

Solution: Always normalize or standardize features before applying k-NN:

- **Min-Max Normalization**: Scale to $[0, 1]$: $x' = \frac{x - \min(x)}{\max(x) - \min(x)}$
- **Standardization**: Scale to mean 0, std 1: $x' = \frac{x - \mu}{\sigma}$

25.7.2 Computational Complexity

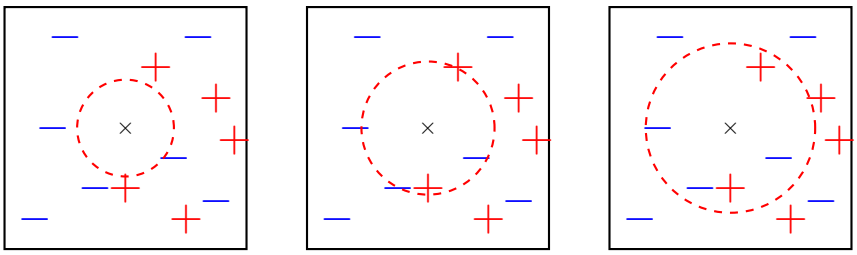
- **Training time**: $O(1)$ (no training, just store data)
- **Prediction time**: $O(Nd)$ where N is training set size, d is dimensionality
- For large datasets, use approximate nearest neighbor algorithms (e.g., KD-trees, Ball trees, LSH)

Note. k-NN is called a **lazy learner** or **instance-based learner** because it doesn't learn a model during training. Instead, it memorizes the training data and performs all computation at prediction time. This makes training fast but prediction slow, especially for large datasets.

25.8 Distance Metrics in k-NN

25.8.1 Euclidean Distance

To calculate the distance, we usually use **Euclidean distance**.



(m) 1-nearest neighbor (m) 2-nearest neighbor (m) 3-nearest neighbor

Euclidean Distance Formula

Given two points $\mathbf{p} = (p_1, p_2, \ldots, p_n)$ and $\mathbf{q} = (q_1, q_2, \ldots, q_n)$:

$$d(\mathbf{p}, \mathbf{q}) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2 + \ldots + (q_n - p_n)^2} = \sqrt{\sum_{i=1}^{n} (q_i - p_i)^2}$$

Observation. The visual shows how the radius grows as k increases:

- **1-NN**: Small circle, includes only the closest neighbor
- **2-NN**: Medium circle, includes 2 closest neighbors
- **3-NN**: Larger circle, includes 3 closest neighbors

The decision boundary depends on which class dominates within the circle.

25.9 Weighted k-NN

25.9.1 The Problem with Equal Weights

All the closest neighbors have the same weights, meaning that they are equally important to make a decision.

Issue: Therefore, the algorithm is very sensitive to the number of k.

25.9.2 Solution: Distance-Based Weighting

This can be reduced by **weighing the contribution of neighbors based on distance** $w = 1/d(\mathbf{x}^*, s_i)$, such that the **closest** neighbor would have more **importance**.

Weighted k-NN

Instead of simple majority voting, each neighbor contributes with weight:

$$w_i = \frac{1}{d(\mathbf{x}^*, s_i)}$$

The class with the highest weighted sum is chosen:

$$\text{class}(\mathbf{x}^*) = \arg\max_c \sum_{i \in U(\mathbf{x}^*)} w_i \cdot \mathbb{1}(y_i = c)$$

where $\mathbb{1}(y_i = c)$ is 1 if neighbor i belongs to class c, and 0 otherwise.

Example 25.3 (Weighted vs. Unweighted k-NN)**.** Consider $k = 3$ with neighbors at distances 1, 2, and 5:

- **Unweighted**: Each neighbor gets weight 1
- **Weighted**: Neighbors get weights $w_1 = 1/1 = 1.0$, $w_2 = 1/2 = 0.5$, $w_3 = 1/5 = 0.2$

The closest neighbor has much more influence in the weighted version.

25.10 Importance of Feature Scaling (Revisited)

25.10.1 Why k is Sensitive

The choice of k is important because:

- If k is too **small**, the approach is **sensitive to noise**
- If k is too **large**, the neighborhood can include examples belonging to **other classes**

Critical Requirement

To operate correctly the attributes must have the **same value scale** and must therefore be **normalized** in the preprocessing phase.

Observation. Without normalization, features with larger ranges will dominate the distance calculation, making features with smaller ranges essentially irrelevant to the classification decision.

25.11 Mahalanobis Distance

25.11.1 When Euclidean Distance is Not Enough

Normalizing the scale may not be sufficient when there are different data distributions.

In such cases instead of using Euclidean distance, one can also try using **Mahalanobis distance**.

25.11.2 Definition

The Mahalanobis distance is a measure of the distance between **a point and a distribution**. It's a generalized form of the Euclidean distance that accounts for **correlations between variables** and **different variances** along different axes.

Mahalanobis Distance Formula

$$D = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}$$

where:

- $\mathbf{x}$ is the n-dimensional vector representing the point's coordinates
- $\boldsymbol{\mu}$ is the n-dimensional vector representing the mean of the distribution
- $\boldsymbol{\Sigma}$ is the $n \times n$ covariance matrix of the distribution
- $\boldsymbol{\Sigma}^{-1}$ is the inverse of the covariance matrix

Observation. When the covariance matrix is the identity matrix (i.e., features are uncorrelated and have unit variance), the Mahalanobis distance reduces to the Euclidean distance.

25.11.3 When to Use Mahalanobis Distance

1. **Correlated Features**: Changes in one feature are associated with changes in another.
2. **Unequal Variances**: MD normalizes the distance based on the variances of the features, preventing features with larger variances from dominating the distance calculation.
3. **Outlier Detection**: Outliers often have a large MD from the rest of the data points, especially if they deviate from the normal covariance pattern.
4. **In high dimensional spaces**: ED can be problematic (*curse of dimensionality*). Incorporating information about the covariance structure, can mitigate this issue to some extent.

Euclidean Distance	**Mahalanobis Distance**
Assumes features are independent	Accounts for correlations between features
Treats all directions equally	Accounts for different variances along different axes
Sensitive to feature scaling	Automatically accounts for different scales
Simple and fast to compute	More computationally expensive (requires covariance matrix)
Works well when features are uncorrelated and have similar variance	Better when features are correlated or have different variances
Distance from point to point	Distance from point to distribution

Table 11: Euclidean vs. Mahalanobis Distance

Example 25.4 (Mahalanobis Distance in Practice)**.** Consider a 2D dataset where:

- Feature 1 has variance 1
- Feature 2 has variance 100
- Features are positively correlated

Euclidean distance would be dominated by Feature 2 due to its large variance.

Mahalanobis distance would:

- Normalize for the different variances
- Account for the correlation structure
- Provide a more meaningful distance measure

Note. Computing Mahalanobis distance requires estimating the covariance matrix $\mathbf{\Sigma}$, which can be challenging when:

- The number of features is large relative to the number of samples
- The covariance matrix is singular (not invertible)
- There is insufficient data to reliably estimate correlations

In such cases, regularization techniques or dimensionality reduction may be needed before applying Mahalanobis distance.

25.11.4 Visual Example: Euclidean vs. Mahalanobis Distance

Example 25.5 (Comparing Distance Metrics)**.** Consider a dataset with a centroid at $(\bar{x}, \bar{y}) = (3.1, 3.0)$ and two data points:

- **Data point 1**: $(5, 5)$ (yellow/orange point)
- **Data point 2**: $(5, 1)$ (blue point)

Using Euclidean Distance:

Distance from centroid to Data point 1:

$$d_1 = \sqrt{(5 - 3.1)^2 + (5 - 3.0)^2} = \sqrt{1.9^2 + 2.0^2} = \sqrt{3.61 + 4.00} = 2.76$$

Distance from centroid to Data point 2:

$$d_2 = \sqrt{(5 - 3.1)^2 + (1 - 3.0)^2} = \sqrt{1.9^2 + (-2.0)^2} = \sqrt{3.61 + 4.00} = 2.76$$

Based on Euclidean distance, both points have the same distance to the centroid!

Observation. However, looking at the scatter plot, we can see that the green points (the cluster) are elongated along a diagonal direction. Data point 1 lies along this diagonal trend, while Data point 2 deviates significantly from it.

Euclidean distance treats all directions equally and doesn't account for the correlation structure of the data.

Example 25.6 (Using Mahalanobis Distance)**.** When we apply Mahalanobis distance to the same scenario:

- $\mathbf{MD_1 = 2.26}$ (Data point 1 to centroid)
- $\mathbf{MD_2 = 6.45}$ (Data point 2 to centroid)

Why is MD_1 so different from MD_2?

Because MD takes into account the **correlation** in the data.

Interpretation:

- Data point 1 follows the natural diagonal trend of the cluster → smaller Mahalanobis distance
- Data point 2 deviates from the correlation pattern → larger Mahalanobis distance

Note. This example illustrates why Mahalanobis distance is superior when features are correlated:

- **Euclidean distance**: Treats both points as equally distant (2.76)

- **Mahalanobis distance**: Correctly identifies that point 2 is much farther from the distribution (6.45 vs 2.26)

Mahalanobis distance accounts for the shape and orientation of the data distribution, not just raw coordinate distances.

25.12 How to Select k in k-NN

Choosing the right value of k is crucial for k-NN performance. Here are practical strategies:

25.12.1 Common Heuristics

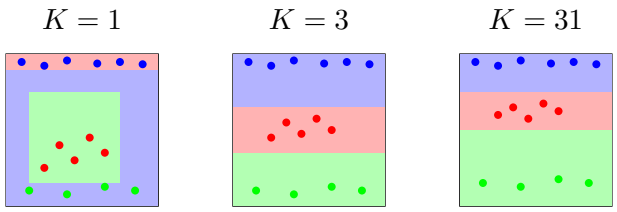
- **Common values**: Often 3, 5, 7
- **Choose an odd number** to avoid ties (*undetermined cases*)
- **Use your training and validation data** to decide k:
  - Change k from $k = 1$ until the validation performance decreases
  - Pay attention not to **underfit** and **overfit** the data
  - Finalize the decision of k and use the same k to classify the test data
- **A typical choice** is $k \approx \sqrt{N}$ where N is the number of training samples

Validation-Based Selection

The most reliable method is to use cross-validation:

1. Try different values of k (e.g., 1, 3, 5, 7, 9, 11, ...)
2. Evaluate performance on validation set for each k
3. Select the k that gives the best validation performance
4. Use this optimal k on the test set

25.12.2 Visual Understanding: k and Decision Boundaries



What is the role of k? How does it relate to overfitting and underfitting?

- $k = 1$ (left): Very complex, jagged decision boundaries
  - High variance, low bias
  - **Overfitting**: Captures noise in the training data
  - Sensitive to outliers
- $k = 3$ (middle): Moderate complexity, smoother boundaries
  - Balanced bias-variance trade-off
  - Good generalization
- $k = 31$ (right): Very smooth, simple decision boundaries
  - Low variance, high bias

  - **Underfitting**: May miss important patterns
  - Too much smoothing

Observation. As k increases:

- Decision boundaries become smoother and simpler
- The model becomes less sensitive to individual data points
- Risk shifts from overfitting (small k) to underfitting (large k)

25.12.3 Handling Ties (Undetermined Cases)

If there are undetermined cases (ties), there are some alternatives:

1. **Arbitrary choice**: Randomly select one of the tied classes
2. **Nearest average sample**: Assign $\mathbf{x}$ to the class (among those "tie") that has the nearest average sample (calculated between the k_i samples)
3. **Least distance**: Assign $\mathbf{x}$ to the class that has the k_i samples with least distance from it

Example 25.7 (Handling a Tie)**.** Suppose $k = 4$ and we have:

- 2 neighbors from class A at distances: 1.5, 2.0
- 2 neighbors from class B at distances: 1.2, 3.5

Solutions:

- **Arbitrary**: Randomly choose A or B
- **Nearest average**:
  - Class A average: $(1.5 + 2.0)/2 = 1.75$
  - Class B average: $(1.2 + 3.5)/2 = 2.35$
  - Choose class A (smaller average distance)
- **Least distance**: Choose class B (has the closest neighbor at 1.2)

Note. The best strategy for handling ties depends on your specific problem:

- **Nearest average**: Good when you want to consider all neighbors equally
- **Least distance**: Good when you believe the closest neighbor is most important
- **Best practice**: Use odd k to avoid ties altogether in binary classification

25.13 k-NN: Pros and Cons

25.13.1 Advantages (Pros)

- **Almost no assumptions about the data**
  - Non-parametric method
  - Doesn't assume any particular distribution
  - Works with any data structure
- **Non-parametric**
  - No model parameters to learn

  - Flexible and adaptive to data

- **There is no training phase, cost is 0**
  - Instant "training" - just store the data
  - Easy to update with new data
  - No convergence issues

- **Allow us to construct non-linear class boundaries and are therefore more flexible**
  - Can capture complex decision boundaries
  - Not limited to linear separability
  - Adapts to local data structure

k-NN gives locally defined decision boundaries between classes

Unlike global methods (e.g., linear classifiers), k-NN creates decision boundaries that adapt to the local structure of the data, allowing for complex, non-linear separations.

Observation. The figure shows how k-NN can create complex, non-linear decision boundaries (shown in red) that follow the natural clustering of the data. The decision boundary is not a simple line or hyperplane, but rather a complex curve that adapts to the local density and distribution of each class (label 1, 2, and 3).

On the right, we see the resulting decision regions where each color represents the area classified as a particular label. The boundaries are highly non-linear and follow the intricate patterns in the data.

25.13.2 Disadvantages (Cons)

- **Need to handle missing data**
  - Cannot compute distances with missing values
  - Requires imputation or special handling

- **The distance metric should be chosen carefully**
  - Different metrics can give very different results
  - Must consider feature correlations and scales
  - May need domain knowledge

- **Sensitive to class outliers** (e.g., mislabeled training instances)
  - Outliers can create incorrect local decision boundaries
  - No mechanism to detect or remove bad training examples

- **Sensitive to many irrelevant attributes** (which influence the distance)
  - All features contribute equally to distance
  - Irrelevant features add noise
  - Feature selection is crucial

- **Computationally expensive:**

- **Poor scalability**: For each new example, we have to perform as many comparisons as there are labeled data
- **It is necessary to store all data points**: Therefore, the method is more appropriate when the number of samples N is low
- Memory requirements grow linearly with dataset size
- Prediction time is $O(Nd)$ where N is training size, d is dimensionality

Scalability Issue

k-NN does not scale well to large datasets. For a dataset with millions of samples, making a single prediction requires computing distances to all training points, which becomes prohibitively expensive.

Note. The main trade-off with k-NN is between flexibility and computational cost:

- **Pros**: Simple, flexible, no training, handles non-linear boundaries
- **Cons**: Slow predictions, high memory, sensitive to irrelevant features and outliers

k-NN works best for:

- Small to medium-sized datasets
- Problems where decision boundaries are complex and non-linear
- Applications where training time is critical but prediction time is less important
- Situations where new data arrives frequently (easy to update)

25.14 How k-NN Was Born: PDF Estimation

k-NN has a theoretical foundation in probability density function (PDF) estimation. Let's explore this connection.

25.14.1 Going Back to PDF Estimation

Recall from our earlier discussion of density estimation:

- Instead of fixing V as in the case of kernels (Parzen windows), we **fix the value of** k and enlarge the radius of the sphere to include k samples
- Let's see the **a-priori probability**:
  - Given a class y_j, the frequency of occurrence of samples N_j of class j is generally simply measured in relation to the total number of samples N, i.e.,

$$P(y_j) \equiv \hat{p}(y_j) = \frac{N_j}{N}$$

Observation. This is the empirical estimate of the prior probability: the proportion of training samples that belong to class j.

25.14.2 Density Estimation and Nearest-Neighbor Rule

Consider 2 classes, y_i and y_j, containing N_i and N_j samples, $N = N_i + N_j$.

The local density estimation for y_i is calculated as (and similarly for y_j):

$$\hat{p}(\mathbf{x}|y_i) = \frac{1}{V}\frac{k_i}{N_i}$$

such that the ratio of k_i (k_j) points to the total N_i (N_j) belonging to the class y_i (y_j) contained in the volume V.

Deriving k-NN from Bayes Rule

The Bayes rule says $p(\mathbf{x}|y_i)P(y_i) > p(\mathbf{x}|y_j)P(y_j)$, then

$$\begin{aligned}
&\hat{p}(\mathbf{x}|y_i)\hat{p}(y_i) > \hat{p}(\mathbf{x}|y_j)\hat{p}(y_j) \\
\Rightarrow &\frac{1}{V}\frac{k_i}{N_i}\cdot\frac{N_i}{N} > \frac{1}{V}\frac{k_j}{N_j}\cdot\frac{N_j}{N} \\
\Rightarrow &\frac{1}{VN_i}k_iN_i > \frac{1}{VN_j}k_jN_j \\
\Rightarrow &k_i > k_j
\end{aligned}$$

The k-NN Rule

Classify $\mathbf{x}$ to class y_i if $k_i > k_j$, i.e., if there are more neighbors from class i than from class j in the volume V around $\mathbf{x}$.

Observation. This derivation shows that k-NN is actually implementing a Bayesian decision rule using local density estimates! The key insight is:

- We estimate $p(\mathbf{x}|y_i)$ locally using the k nearest neighbors
- We estimate $P(y_i)$ using the proportion of class i in the training set
- The Bayes rule simplifies to: choose the class with the most neighbors in the local region

Note. This theoretical foundation explains why k-NN works:

- It's approximating the optimal Bayes classifier
- It uses local density information to make decisions
- The quality of the approximation depends on:
  - Choice of k (too small = noisy estimates, too large = oversmoothing)
  - Density of training data (more data = better estimates)
  - Dimensionality (curse of dimensionality affects density estimation)

As the number of training samples $N \to \infty$ and $k \to \infty$ (but $k/N \to 0$), the k-NN classifier approaches the optimal Bayes classifier.

Part IV

Decision Trees and Ensemble Methods

26 Introduction to Decision Trees

Decision trees are fundamental machine learning models that provide both powerful predictive capabilities and high interpretability. They are tree-structured prediction models that make decisions by routing data through a hierarchy of tests.

26.1 Decision Tree Structure

A decision tree is composed of two types of nodes:

- **Non-terminal nodes**: These nodes have 2 or more children and implement a **routing function**. They perform tests on the input features to decide which branch to follow.
- **Leaf nodes (terminal nodes)**: These nodes have no children and implement a **prediction function**. They provide the final output of the model.

Key Structural Properties

- There are no cycles in the tree structure
- All nodes have at most 1 parent
- There is exactly one node with no parent, called the **root node**
- The tree forms a directed acyclic graph (DAG)

26.2 How Decision Trees Work

A decision tree takes an input $\mathbf{x} \in \mathcal{X}$ and routes it through its nodes until it reaches a leaf node, where a prediction is made.

The process:

1. The input enters at the root node
2. Each node represents a test on a variable (feature)
3. An edge to a child node represents the outcome of the test
4. The routing continues until a leaf node is reached
5. At the leaf, a prediction is made based on the prediction function

26.3 Decision Tree Inference: Non-terminal Nodes

Each non-terminal node can be formally represented as $\text{Node}(\phi, t_L, t_R)$, where:

- $\phi : \mathcal{X} \to \{L, R\}$ is a **routing function**
- t_L is the left child subtree
- t_R is the right child subtree

When an input $\mathbf{x}$ reaches the node:

- If $\phi(\mathbf{x}) = L$, the input goes to the left child t_L

- If $\phi(\mathbf{x}) = R$, the input goes to the right child t_R

Note: We assume binary trees here, but the concept generalizes to trees with more than two children.

26.4 Decision Tree Inference: Leaf Nodes

Each leaf node can be represented as Leaf(h), where:

- $h \in \mathcal{F}_{\text{task}}$ is a **prediction function** (typically a constant)
- Depending on the task:
  - For **classification**: $h \in \mathcal{Y}$ (a class label)
  - For **regression**: $h \in \mathbb{R}$ (a continuous value)

Once $\mathbf{x}$ reaches the leaf, the final prediction is given by $h(\mathbf{x})$.

26.5 Formal Decision Tree Function

The complete decision tree function can be defined recursively:

$$f_t(\mathbf{x}) = \begin{cases} h(\mathbf{x}) & \text{if } t = \text{Leaf}(h) \\ f_{t_{\phi(\mathbf{x})}}(\mathbf{x}) & \text{if } t = \text{Node}(\phi, t_L, t_R) \end{cases}$$

Interpretation:

- If we are at a leaf node, return the prediction $h(\mathbf{x})$
- If we are at a non-terminal node, apply the routing function $\phi(\mathbf{x})$ to determine which child to visit, then recursively apply the tree function to that child

26.6 Decision Tree Inference Example

Consider a 2D classification problem with classes $\xi \in \{+, \times, \circ, \triangle, \square\}$. We can build a decision tree with routing functions:

$$\phi_1(x, y) = \begin{cases} L & \text{if } y \geq 3 \\ R & \text{else} \end{cases}$$

$$\phi_2(x, y) = \begin{cases} L & \text{if } x \leq 3 \\ R & \text{else} \end{cases}$$

$$\phi_3(x, y) = \begin{cases} L & \text{if } x \leq 2 \\ R & \text{else} \end{cases}$$

$$\phi_4(x, y) = \begin{cases} L & \text{if } x \leq 4 \\ R & \text{else} \end{cases}$$

These routing functions partition the 2D space into axis-aligned rectangular regions, with each region corresponding to a leaf node that predicts a specific class.

27 Learning Decision Trees

27.1 The Learning Problem

Given a training set $\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$, we want to find a tree $t^* \in \mathcal{T}$ from the set of all possible decision trees $\mathcal{T}$ that best fits the data.

The Challenge

This optimization problem has many solutions and can be **NP-hard** in general. Thus, we need to impose constraints, such as:

- Finding the most compact tree
- Using greedy algorithms that find locally optimal splits
- Limiting tree depth or complexity

27.2 Tree Learning Framework

To learn a decision tree, we need to specify:

1. $\mathcal{H}_{\textbf{leaf}}$: A set of leaf prediction functions (e.g., constant functions)
2. Φ: A set of possible splitting functions $\Phi \subseteq \{L, R\}^{\mathcal{X}}$
3. **Tree-growing strategy**: A recursive algorithm that partitions the training set and decides whether to grow leaves or non-terminal nodes

Two Popular Algorithms:

- **ID3** (Iterative Dichotomiser 3): Developed by Ross Quinlan
- **CART** (Classification and Regression Trees): Developed by Breiman et al.

27.3 Tree Growing Strategy

The tree-growing process works recursively:

1. Start with all training examples at the root
2. Find the best splitting function $\phi \in \Phi$ that partitions the data
3. For each resulting subset:
    - If the subset is pure or meets stopping criteria, create a leaf node
    - Otherwise, recursively apply the splitting process

28 ID3 and CART Algorithms

28.1 Example: Playing Tennis Dataset

Consider a classic example with 14 training samples and 4 attributes:

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

- 14 training samples
- 4 attributes (features)
- 9 samples with class "Yes"
- 5 samples with class "No"

28.2 Choosing the Best Attribute

The fundamental question in building a decision tree is: **How do we pick the best attribute to split on?**

Consider two possible splits on the root node:

Split 1: Outlook attribute

- Sunny: 2 yes / 3 no
- Overcast: 4 yes / 0 no (pure!)
- Rain: 3 yes / 2 no

Split 2: Wind attribute

- Weak: 6 yes / 2 no
- Strong: 3 yes / 3 no

Which split is better?

The **Outlook** split is better because:

- It produces one pure subset (Overcast: 4 yes / 0 no)
- Pure subsets mean we are completely certain about the classification
- The Weak wind subset (6 yes / 2 no) is better than Strong (3 yes / 3 no), but neither is pure

28.3 Measuring Purity

We need a formal way to measure the "purity" of a split. Key properties:

- **Pure set** (4 yes / 0 no): Completely certain (100% confidence)
- **Impure set** (3 yes / 3 no): Completely uncertain (50% confidence)

- The measure should be **symmetric**: 4 yes / 0 no is as pure as 0 yes / 4 no

There are different ways to measure purity, with **entropy** being the most popular.

29 Entropy and Information Gain

29.1 Entropy

For binary classification problems with positive and negative classes:

- $p(+)$ is the proportion of positives in a subset
- $p(-)$ is the proportion of negatives in a subset

Entropy Formula

The entropy of a subset S is:

$$H(S) = -p(+)\log_2 p(+) - p(-)\log_2 p(-)$$

Important properties:

- Entropy measures uncertainty or impurity
- Higher entropy = more impure = more uncertain
- Lower entropy = more pure = more certain

29.1.1 Entropy Examples

Impure subset (3 yes / 3 no):

$$\begin{aligned} p(+) &= 0.5, \quad p(-) = 0.5 \\ H(S) &= -0.5\log_2(0.5) - 0.5\log_2(0.5) \\ &= -0.5 \times (-1) - 0.5 \times (-1) \\ &= 0.5 + 0.5 = 1.0 \end{aligned}$$

Pure subset (4 yes / 0 no):

$$\begin{aligned} p(+) &= 1.0, \quad p(-) = 0.0 \\ H(S) &= -1.0\log_2(1.0) - 0.0\log_2(0.0) \\ &= -1.0 \times 0 - 0 \\ &= 0.0 \end{aligned}$$

Interpretation:

- Maximum entropy (1.0) for the impure set (50/50 split)
- Minimum entropy (0.0) for the pure set (100% one class)
- Higher numbers correspond to less pure subsets!

29.2 Information Gain

We want to have many items in pure sets after splitting. We calculate the **expected drop in entropy** after each split.

Information Gain Formula

The information gain of attribute A on set S is:

$$\text{Gain}(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

Where:

- $H(S)$ is the entropy of the original set
- Values(A) are the possible values of attribute A
- S_v is the subset of S where $x_A = v$
- $|S_v|/|S|$ weights each subset by its size

Interpretation:

- Information gain is the **mutual information** between attribute A and class labels
- It measures how much uncertainty is reduced by knowing attribute A
- We want to **maximize information gain**
- Higher gain = better attribute for splitting

29.3 Using Information Gain for Attribute Selection

The ID3 algorithm uses information gain to decide which attribute to pick:

1. For every attribute in your data, compute the information gain
2. Select the attribute with the **highest information gain**
3. This attribute reduces uncertainty the most and leads to the purest possible split

Note: We consider one level split at a time, but the procedure is recursive. After splitting on an attribute, we recursively apply the same process to each resulting subset.

29.4 Problems with Information Gain

Information gain has a significant problem: it favors attributes with many values.

Example: Consider the "Day" attribute (D1, D2, ..., D14) in the tennis dataset:

- Each day is unique, so splitting on "Day" creates 14 subsets
- Each subset contains exactly one example
- All subsets are perfectly pure (optimal split according to information gain!)
- But this is useless! It won't generalize to new data

If we encounter a new example "D15: Rain, High, Weak", the tree cannot classify it because it only knows about days D1-D14.

The problem: Information gain can lead to overfitting by favoring attributes that perfectly separate the training data but generalize poorly to test data.

29.5 Gain Ratio: A Solution

One solution is to use **Gain Ratio** instead of information gain:

$$\text{GainRatio}(S, A) = \frac{\text{Gain}(S, A)}{\text{SplitInfo}(S, A)}$$

Where:

$$\text{SplitInfo}(S, A) = -\sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \log_2 \frac{|S_v|}{|S|}$$

Effect:

- SplitInfo measures how broadly and uniformly the attribute splits the data
- Dividing by SplitInfo **penalizes attributes with many values**
- This helps prevent overfitting to attributes like "Day"

29.6 Advantages of Information Gain

Despite its limitations, information gain (and decision trees in general) has important advantages:

- **Interpretability**: Decision trees are easy to understand
- **Readable rules**: It's possible to extract concise rules describing classification decisions
- **No "black box"**: The decision-making process is transparent
- **Important for users**: Explainability is crucial in many applications (medical diagnosis, loan approval, etc.)

Example rule from a decision tree:

```
IF Outlook = Overcast THEN PlayTennis = Yes
IF Outlook = Sunny AND Humidity = High THEN PlayTennis = No
```

30 Alternative Splitting Criteria

Besides entropy and information gain, there are other measures for selecting splits.

30.1 Gini Impurity

The **CART algorithm** uses Gini impurity instead of entropy.

Gini Impurity Formula

For a set S with C classes:

$$\text{Gini}(S) = 1 - \sum_{i=1}^{C} p_i^2$$

Where p_i is the proportion of samples belonging to class i.

Properties:

- Gini impurity = 0 for a pure set (all samples from one class)
- Gini impurity = 0.5 for a binary classification with 50/50 split

- When building a decision tree, we choose features with the **least Gini index** as the root node

30.2 Comparison of Splitting Criteria

Criterion	Task	Formula	Description
Entropy	Classification	$-\sum_{i=1}^{C} p_i \log_2(p_i)$	p_i is the frequency of label i at a node, C is the number of unique labels. Maximize the Gain.
Gini Impurity	Classification	$1 - \sum_{i=1}^{C} p_i^2$	p_i is the frequency of label i at a node, C is the number of unique labels. Minimize the Gini.
Variance	Regression	$\frac{1}{N}\sum_{i=1}^{N}(y_i - \mu)^2$	y_i is the label for instance i, N is the number of instances, $\mu = \frac{1}{N}\sum_{i=1}^{N} y_i$. Minimize the variance.

31 Decision Trees and Overfitting

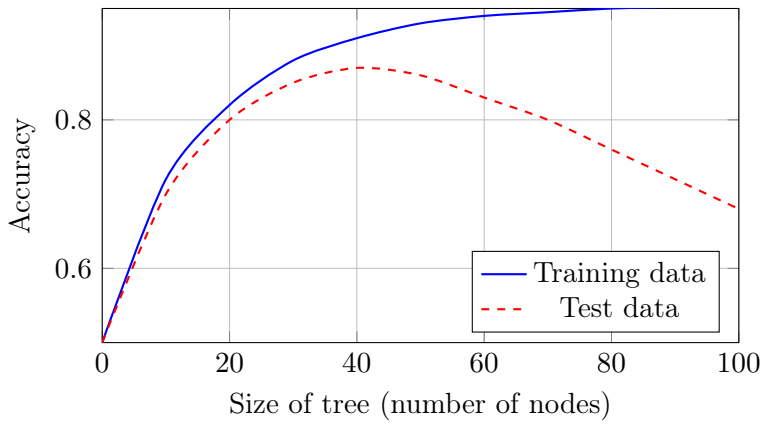
31.1 Why Decision Trees Overfit

Decision trees are **non-parametric models** with a structure determined by the data. As a result:

- They are very **flexible** and can easily fit the training set
- They have a **high risk of overfitting**
- Without constraints, they can create a leaf for every training example

31.2 Overfitting Example

Consider the relationship between tree size and accuracy:



Observations:

- Training accuracy continues to increase with tree size
- Test accuracy initially increases, then decreases
- The optimal tree size is around 30-40 nodes
- Larger trees overfit to the training data

31.3 Techniques to Prevent Overfitting

Standard machine learning techniques apply to decision trees:

1. **Early stopping**: Stop growing the tree before it perfectly fits the training data
   - Set a maximum depth
   - Require a minimum number of samples to split
   - Require a minimum information gain to split
2. **Pruning**: Build a full tree, then remove branches that don't improve validation performance
3. **Regularization**: Add penalties for tree complexity
4. **Data augmentation**: Increase training data
5. **Ensemble methods**: Combine multiple trees (Random Forest, boosting)

31.4 Pruning (Post-processing)

Pruning is a technique to reduce complexity *after* tree building:

1. **Build the full tree**: Grow the tree until each leaf node is pure or a stopping criterion is met
2. **Evaluate nodes from bottom to top**: Starting from the bottom, evaluate each node to see if pruning it would improve performance on a validation set
3. **Define cost-complexity measure**: Consider both the accuracy of the node and the size of the subtree rooted at that node
4. **Prune nodes iteratively**: Identify the node with the smallest cost-complexity measure and prune its subtree (turn it into a leaf labeled with the majority class)
5. **Repeat until done**: Continue until a suitable tree is achieved or further pruning doesn't improve performance

32 Decision Trees for Different Data Types

32.1 Continuous Attributes

So far, we have focused on categorical attributes, but decision trees can also handle continuous attributes.

Strategy: Pick a threshold to create a binary split.

For example, for a continuous attribute like temperature:

$$\phi(\text{temperature}) = \begin{cases} L & \text{if temperature} > 77.8 \\ R & \text{else} \end{cases}$$

How to choose the threshold?

- Consider all possible split points (or a sample of them)
- For each candidate threshold, compute the information gain (or Gini impurity reduction)
- Choose the threshold that maximizes information gain

Geometric interpretation:

- Each split creates an axis-aligned boundary in feature space
- For 2D data with continuous features x_1 and x_2, splits create rectangular regions
- Example splits: $x_1 > \theta_1$, $x_2 < \theta_2$, $x_2 > \theta_3$, $x_1 < \theta_4$

32.2 Multi-class Classification

Decision trees naturally extend to multi-class problems (k different classes):

- At each leaf, predict the **most frequent class** in that subset
- Entropy for multi-class:

$$H(S) = -\sum_{c=1}^{k} p(c) \log_2 p(c)$$

  Where $p(c)$ is the proportion of examples of class c in subset S
- Information gain and Gini impurity formulas extend naturally to k classes

33 Decision Trees: Summary

33.1 Advantages (Pros)

- **Interpretable**: Humans can understand the reason for decisions
- **Handles irrelevant data**: Irrelevant features have Gain = 0 and won't be selected
- **Handles missing data**: Can be adapted to deal with missing values
- **Very compact**: After pruning, # nodes $\ll$ # training data
- **Fast at test time**: O(depth), where depth $\ll$ # training data
- **No feature scaling needed**: Works with features on different scales

33.2 Disadvantages (Cons)

- **Greedy algorithm**: May not find the globally optimal tree (locally optimal choices at each split)
- **NP-hard problem**: Finding the optimal tree is exponentially difficult (exponentially many possible trees)
- **Only axis-aligned splits**: Cannot naturally capture diagonal decision boundaries
  - Linear classifiers can have diagonal boundaries
  - Decision trees create rectangular partitions
  - This limits expressiveness for some problems
- **High variance**: Small changes in training data can lead to very different trees
- **Prone to overfitting**: Especially with deep trees

33.3 Key Takeaways

- **ID3 algorithm**: Grows decision tree from root down, greedily selecting the next best attribute using information gain
- **Entropy**: Measures uncertainty (impurity) in a set
- **Information Gain**: Measures the reduction in uncertainty following a split
- **Complete hypothesis space**: Searches the entire space of possible trees
- **Preference**: Prefers smaller trees and high information gain at the root
- **Overfitting**: Addressed by post-pruning (remove nodes while accuracy increases on validation set)
- **Performance**: Fast, compact, and interpretable

34 Random Forests

Random forests address many limitations of individual decision trees by combining multiple trees into an ensemble.

34.1 What is a Random Forest?

Random forests are ensembles of decision trees.

- Each tree is trained on a **bootstrapped version** of the training set (sampled with replacement)
- This creates diversity among the trees
- Different trees make different errors
- Combining their predictions reduces overall error

34.1.1 Sampling with Replacement (Bootstrapping)

Given a dataset $\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$:

1. For each tree t_i, create a new dataset $\mathcal{D}_i$ by:
   - Randomly sampling n examples from $\mathcal{D}$ **with replacement**
   - Some examples may appear multiple times
   - Some examples may not appear at all (approximately 37% are left out)
2. Train tree t_i on dataset $\mathcal{D}_i$
3. Repeat for all T trees in the forest

34.2 Random Feature Selection

In addition to bootstrap sampling, random forests introduce randomness in feature selection:

- At each node, instead of considering all features for splitting, only a **random subset** of features is considered
- Typical choices:
  - For classification: $\sqrt{d}$ features, where d is the total number of features

  - For regression: $d/3$ features

- This helps to **decorrelate** decision trees
- **Extremely randomized trees** take this further by sampling split thresholds randomly instead of optimizing them

34.3 Random Forest Prediction

The final prediction of a random forest is obtained by **aggregating** the predictions of all trees:

Random Forest Prediction

For classification:

$$\hat{y}_{\text{RF}}(\mathbf{x}) = \text{mode}\{f_{t_1}(\mathbf{x}), \ldots, f_{t_T}(\mathbf{x})\}$$

(Majority vote across all trees)

For regression:

$$\hat{y}_{\text{RF}}(\mathbf{x}) = \frac{1}{T}\sum_{i=1}^{T} f_{t_i}(\mathbf{x})$$

(Average of all tree predictions)

34.4 Why Random Forests Work

Random forests work well because of several key principles:

1. **Variance reduction**: Averaging multiple models reduces variance without increasing bias
2. **Decorrelation**: Random feature selection ensures trees make different types of errors
3. **Robustness**: Errors of individual trees tend to cancel out
4. **Overfitting resistance**: Individual trees may overfit, but the ensemble typically does not

Bias-Variance Trade-off:

- Individual deep trees have low bias but high variance
- Averaging many trees keeps bias low while reducing variance
- Result: Better generalization performance

35 Ensemble Methods

Random forests are one example of a broader class of techniques called **ensemble methods**.

35.1 What are Ensemble Methods?

Ensemble learning combines different models to obtain more accurate predictions than any individual model.

Ensemble Learning Principle

"Wisdom of crowds": Aggregating predictions from multiple models often yields better performance than using a single model.

35.2 Types of Ensemble Methods

There are three main types of ensemble methods:

1. **Bagging** (Bootstrap Aggregating)
2. **Boosting**
3. **Stacking**

36 Bagging (Bootstrap Aggregating)

36.1 The Bagging Algorithm

Bagging is a technique where we train T models on T different samplings of the dataset.

Procedure:

1. Take the original dataset $\mathcal{D}$
2. Create T bootstrap samples:
   - $\mathcal{D}_1$: Sample n examples from $\mathcal{D}$ with replacement
   - $\mathcal{D}_2$: Sample n examples from $\mathcal{D}$ with replacement
   - ...
   - $\mathcal{D}_T$: Sample n examples from $\mathcal{D}$ with replacement
3. Train a model on each bootstrap sample:
   - Model 1 trained on $\mathcal{D}_1$
   - Model 2 trained on $\mathcal{D}_2$
   - ...
   - Model T trained on $\mathcal{D}_T$
4. **Aggregate** predictions:
   - Classification: Majority vote
   - Regression: Average

36.2 Benefits of Bagging

- **Reduces variance**: Different bootstrap samples lead to different models; averaging reduces variance
- **Doesn't increase bias**: Each model is trained on a dataset of the same size as the original
- **Robust to overfitting**: Individual models may overfit to their bootstrap samples, but averaging helps
- **Parallel training**: All models can be trained independently and in parallel

36.3 When to Use Bagging

Bagging works best with:

- **High-variance, low-bias models**: e.g., deep decision trees

- Models that are sensitive to small changes in training data
- Situations where you want to reduce overfitting

37 Boosting

Boosting is a different ensemble strategy that trains models **sequentially**, with each new model focusing on the errors of previous models.

37.1 Key Differences from Bagging

Aspect	**Bagging**	**Boosting**
Training	Parallel (independent)	Sequential (dependent)
Sampling	Uniform (with replacement)	Weighted (focus on errors)
Model weights	Equal	Based on performance
Goal	Reduce variance	Reduce bias and variance

37.2 Boosting Intuition

- Trained models are **weighted based on their performance**
- Weak classifiers are learned **sequentially**
- The more mistakes the previous classifier makes on a dataset, the more those misclassified examples are **weighted for the next classifier**
- Each new model focuses on the "hard" examples that previous models got wrong

37.3 Weak Learners

Ensemble methods often use **weak learners** rather than complex models.

Definition 37.1 (Weak Learner)**.** A weak learner is a classification model that performs **slightly better than random guessing**.

For binary classification with balanced classes:

- Random guessing: 50% accuracy
- Weak learner: slightly better than 50% (e.g., 55%)

Examples of weak learners:

- **Decision stump**: A decision tree with only one split (depth = 1)
- **1-Nearest Neighbor**: k-NN with k=1 on a subset of features
- Simple linear classifiers on a single feature

Why use weak learners?

- Fast to train
- Less likely to overfit individually
- Ensemble of many weak learners can create a strong classifier
- "Two heads are better than one" principle

37.4 AdaBoost (Adaptive Boosting)

AdaBoost is one of the most popular boosting algorithms.

AdaBoost Algorithm

Step 1: Initialize weights uniformly for all training samples:

$$w_i^{(1)} = \frac{1}{n} \quad \text{for } i = 1, \ldots, n$$

Step 2: For $t = 1$ to T:

1. Train a weak classifier h_t using the current weights $w^{(t)}$
2. Compute the weighted error:

$$\epsilon_t = \sum_{i:h_t(\mathbf{x}_i) \neq y_i} w_i^{(t)}$$

3. Compute classifier weight:

$$\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$$

4. Update sample weights:

$$w_i^{(t+1)} = w_i^{(t)} \cdot \exp\left(-\alpha_t y_i h_t(\mathbf{x}_i)\right)$$

(Increase weight for misclassified samples, decrease for correctly classified)

5. Normalize weights: $w^{(t+1)} \leftarrow w^{(t+1)} / \sum_i w_i^{(t+1)}$

Step 3: Final classifier:

$$H(\mathbf{x}) = \text{sign} \left(\sum_{t=1}^{T} \alpha_t h_t(\mathbf{x}) \right)$$

(Weighted vote of all weak classifiers)

37.5 AdaBoost: Key Steps Explained

Step 1: Initialize with equal weights

- All training samples start with equal importance
- Build the first weak classifier on this uniformly weighted dataset
- Check which samples are correctly classified and which are not

Step 2: Increase weights for misclassified samples

- Samples that were misclassified get higher weights
- This makes classifying them correctly more important for the next model
- The next weak classifier will focus more on these "hard" examples

Step 3: Iterate until convergence

- Repeat until all data points are correctly classified
- OR until a maximum number of iterations is reached
- Each iteration focuses on the remaining difficult examples

37.6 AdaBoost Properties

- **Adaptive**: Adapts to the errors of previous weak learners
- **Reduces bias and variance**: Combines weak learners to create a strong learner
- **Sensitive to noisy data**: Can overfit if there are many outliers (keeps increasing their weights)
- **No need for feature selection**: Weak learners can be based on individual features
- **Fast training**: Each weak learner is simple and fast to train

38 Stacking

Stacking (stacked generalization) is a more sophisticated ensemble method that uses a **meta-model** to combine base models.

38.1 What is Stacking?

Stacking combines predictions of multiple ML models (called **base models**) to improve overall performance.

Key differences from bagging and boosting:

- Base models are typically **heterogeneous** (different types of models)
- Uses a **meta-model** to learn how to combine base model predictions
- Can be used for both regression and classification
- Often more powerful than bagging or boosting alone

38.2 The Stacking Process

Stacking Algorithm

Level 0 (Base Models):

1. Train multiple diverse base models on the training set:
   - Model 1: e.g., Decision Tree
   - Model 2: e.g., Logistic Regression
   - Model 3: e.g., Support Vector Machine
   - Model 4: e.g., Neural Network
2. Generate predictions from each base model on the training set

Level 1 (Meta-Model):

1. Use the predictions from base models as **features**
2. Train a **meta-model** that learns how to combine these predictions
3. The meta-model learns which base models to trust for which types of inputs

Prediction:

1. For a new input $\mathbf{x}$:
   - Get predictions from all base models
   - Feed these predictions to the meta-model
   - Meta-model outputs the final prediction

38.3 Why Stacking Works

- **Diversity**: Different models capture different patterns in the data
- **Complementary strengths**: Each model may excel in different regions of the input space
- **Learned combination**: The meta-model learns the optimal way to combine predictions (better than simple averaging)
- **Flexibility**: Can combine any types of models

38.4 Stacking Best Practices

- **Use diverse base models**: Different model types, different hyperparameters
- **Avoid data leakage**: Use cross-validation to generate training predictions for the meta-model
- **Keep meta-model simple**: Often a simple linear model or logistic regression works well
- **Regularization**: Add regularization to prevent the meta-model from overfitting

38.5 Comparison of Ensemble Methods

Method	Base Models	Training	Combination
Bagging	Same type (homogeneous), trained on bootstrap samples	Parallel, independent	Equal weight voting/averaging
Boosting	Same type (weak learners), trained sequentially	Sequential, each focuses on previous errors	Weighted voting based on model performance
Stacking	Different types (heterogeneous)	Parallel for base models, then meta-model	Meta-model learns optimal combination

38.6 Key Takeaways

Main Lessons

1. **Single models vs. Ensembles**: Ensembles generally outperform single models
2. **Bias-Variance tradeoff**: Different ensemble methods address different aspects
   - Bagging reduces variance
   - Boosting reduces both bias and variance
   - Stacking leverages diverse model strengths
3. **Interpretability vs. Performance**: Trade-off between understanding and accuracy
4. **Computational cost**: Ensembles are slower to train and predict than single models
5. **Real-world impact**: Random forests and boosting (e.g., XGBoost, LightGBM) are among the most successful ML methods in practice